

A CYCLUS FRAMEWORK FOR ACCELERATOR-DRIVEN TRANSMUTATION OF
THE U.S. SPENT NUCLEAR FUEL INVENTORY

BY

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THESIS

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Abstract

The transmutation of minor actinides in an accelerator-driven subcritical system (ADS) is often proposed as a means of reducing the long-term radiological hazard of used nuclear fuel. Evaluating such a mission at the national scale requires connecting a realistic description of the existing spent-fuel inventory to a model of the burner that would consume it. This thesis develops two facility archetypes for the Cyclus fuel cycle simulator to make that connection. The first, **us_inventory**, represents the U.S. inventory of used fuel at the resolution of individual assemblies, drawn from the STANDARDS 5.0.1 dataset, and supplies material to downstream facilities according to a choice of ranking policies while preserving each assembly’s composition and geographic provenance. The second, **ads**, models an accelerator-driven minor-actinide burner using precomputed Serpent depletion cases as a reduced-order model, transforming a loaded core into its irradiated composition and recording the actinide mass destroyed each cycle.

Using these archetypes, a transmutation fuel cycle is simulated for two burner configurations—a minor-actinide-only core and a fifty-fifty plutonium–minor-actinide core—over 30- and 100-year horizons in once-through and recycling modes. The results are governed by the scarcity of the feed: the accessible minor-actinide inventory is only about 104 t, fewer than four loadings of the large minor-actinide core. As a result, the minor-actinide burner is feed-limited, stalling after three cores without recycling and five with it, whereas the smaller fifty-fifty core, sustained by the far more abundant plutonium, runs continuously across the horizon. The transmutation mission is therefore constrained by how quickly its dilute feed can be retrieved, separated, and recycled rather than by installed reactor power, and because minor actinide is only about 0.1% of the spent fuel, the mission changes the isotopic character of the waste without reducing the mass sent to disposal. The archetypes and the assembly-resolved, provenance-preserving results they produce provide a foundation for subsequent radiological, economic, and siting analyses of accelerator-driven transmutation.

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τ	Time taken to drink one cup of coffee.
μg	Micrograms (of caffeine, generally).

Chapter 1

Introduction

Nuclear power has supplied the United States with large quantities of low-carbon electricity for more than six decades. The fuel discharged from those reactors, however, remains a long-term national responsibility. By the end of 2022, commercial reactors in the United States had permanently discharged roughly 91,000 metric tons of uranium (MTU) of used nuclear fuel (UNF), and the inventory continues to grow by approximately 2,000 MTU each year [1]. This material is distributed across more than one hundred sites in thirty-nine states, including operating nuclear power plants, decommissioned reactor sites, and independent spent fuel storage installations. Because the United States does not yet operate a permanent geologic repository for commercial spent fuel, most of the inventory remains stored at or near the sites where it was generated [2].

The challenge is not simply that the inventory is large. It is also highly heterogeneous. The phrase “the U.S. used nuclear fuel inventory” can suggest a single stockpile with one representative composition, but the actual inventory is the accumulated result of decades of reactor operation. Assemblies discharged in different years were fabricated with different initial enrichments, irradiated to different burnups, operated in different reactor types, and cooled for different lengths of time. Their remaining heavy-metal masses and isotopic compositions therefore vary considerably. Some assemblies contain larger quantities of residual fissile material, plutonium, or minor actinides than others. Some have cooled for several decades, while others were discharged recently and continue to generate substantial decay heat. Any strategy that attempts to process, recycle, or transmute this material must eventually confront those differences.

UNF also changes with time. During the first decades after discharge, its heat generation and radiation field are strongly influenced by fission products. As those nuclides decay, the relative importance of plutonium and the minor actinides increases. Their long half-lives contribute to the need to isolate untreated fuel over timescales on the order of 10^5 years [3]. Among the long-lived components of used fuel, the actinides are distinctive because they can be destroyed through fission rather than managed only through continued storage and radioactive decay. That possibility does not eliminate the need for a repository, nor does it remove every radioactive waste stream. It does, however, create the possibility of changing the quantity and composition of the material that must ultimately be isolated.

Accelerator-driven system (ADS) have been proposed as one technology for carrying out this transmutation. An ADS couples an externally generated neutron source to a subcritical multiplying core. Because the core remains below criticality, its operation depends on the external source, allowing fuel compositions with high concentrations of minor actinides to be considered under conditions that would be more difficult to accommodate in a critical reactor. Detailed studies have examined accelerator reliability, spallation targets,

subcritical-core physics, liquid-metal coolants, fuel behavior, and the rates at which plutonium and minor actinides could be consumed. These studies establish the physical basis for accelerator-driven transmutation and provide increasingly realistic designs for future facilities.

A technically feasible core, however, is not the same as a viable transmutation strategy. A reactor-physics calculation may show how much actinide material one system can consume, but it does not determine where that material will come from, which assemblies will be processed first, how the feed will change over time, or how many facilities will be required. Before an ADS can consume actinides, spent fuel must be selected from storage, transported, cooled as needed, separated, and converted into a suitable feed form. The transmutation facility must then be deployed, supplied, operated, and eventually integrated with storage, recycle, and waste-management facilities. The material discharged from the ADS may require further processing or repeated recycle before the desired degree of actinide destruction is achieved.

The central problem is therefore a fuel-cycle problem rather than only a single-core problem. It requires a model that can follow material from the existing inventory, through selection and processing, into a transmutation facility, and finally into the inventories and waste streams that remain. Such a model must operate over timescales of years or decades while retaining enough information to distinguish among assemblies. It must also allow facilities to compete for resources, enforce throughput limits, and respond to changes in material availability. These capabilities are difficult to represent using a single average spent-fuel composition.

Many previous fuel-cycle studies have necessarily simplified the inventory by representing it as an aggregated material stream or a small set of representative recipes. That approach is useful when comparing generic future fuel cycles or studying equilibrium reactor fleets. It is less satisfactory when the starting point is the existing U.S. inventory. A national-average recipe cannot identify whether the material it represents comes from older or newer assemblies, whether it has high or low burnup, whether it is concentrated at a few sites or dispersed across the country, or which assemblies would actually be selected under a particular management strategy. These differences may affect the amount and quality of material available for transmutation and the characteristics of the inventory that remains untreated.

Recent inventory databases make a more realistic representation possible. The STANDARDS 5.0.1 database contains assembly-level information describing the U.S. commercial spent-fuel inventory, including attributes such as discharge date, burnup, initial enrichment, mass, and calculated isotopic composition. This level of detail creates an opportunity to treat the national inventory not as a single homogeneous stockpile, but as a collection of discrete resources that can be selected according to explicit criteria. The database alone, however, does not describe how those assemblies would move through a future fuel cycle. The information must be incorporated into a dynamic simulator capable of representing transactions among facilities.

The work is conducted within the Advanced Reactors and Fuel Cycles group at University of Illinois Urbana-Champaign (UIUC), where agent-based fuel-cycle simulation is used to examine the system-level consequences of advanced nuclear technologies. It also contributes to the UIUC-led Enhanced Integrated Nuclear Systems for Transmutation and Efficient Isolation of Nuclides (EINSTEIN) project under the Advanced Research Projects Agency-Energy (ARPA-E) Nuclear Energy Waste Transmutation Optimized Now (NEWTON) program, which brings together neutronics, nuclear-data, materials, fuel-cycle, and techno-economic analyses to evaluate externally driven transmutation systems. My own research in the group has involved working directly with CYCLUS, detailed spent-fuel inventory data, and higher-fidelity reactor calculations, which made the disconnect between the available inventory information and existing system-level transmutation models especially clear. That combination of experience placed me in a position to build the missing connection between the fuel assemblies that exist today and the transmutation technologies being

proposed for the future.

This thesis uses the CYCLUS nuclear fuel cycle simulation framework to make that connection. CYCLUS represents the fuel cycle as a system of interacting agents, including regions, institutions, and facilities. Individual facilities request, offer, and exchange material through the Dynamic Resource Exchange, while modular software components known as archetypes define their behavior. This structure makes it possible to represent the U.S. inventory as an active supplier rather than as a static input file. It also allows that supplier to interact with processing, storage, and transmutation facilities within the same simulation.

Two new facility archetypes are developed for this purpose. The `us_inventory` archetype represents individual U.S. spent-fuel assemblies as selectable material resources. Each assembly retains metadata including its heavy-metal mass, discharge date, burnup, initial enrichment, and isotopic composition derived from Storage Transportation and Disposal Analysis Resource and Data System. (STANDARDS) 5.0.1 [4], [5]. Rather than collapsing these records into one average composition, the facility supplies material through the CYCLUS Dynamic Resource Exchange according to user-defined throughput limits and inventory-selection policies.

The second archetype represents an ADS. Its material transformation is based on precomputed Serpent depletion calculations, allowing the fuel-cycle simulation to incorporate higher-fidelity neutronics results without performing a full transport and depletion calculation during every simulation time step. Coupling the two archetypes makes it possible to follow material from individual inventory records into the modeled transmutation system and to observe how the remaining inventory changes as material is withdrawn.

The selection policies considered in this work rank available assemblies according to characteristics such as discharge date, burnup, and initial enrichment. These policies are not intended to represent final operational decisions or recommendations for U.S. waste management. Instead, they provide controlled cases for examining how different assumptions about access to the inventory affect the material supplied to a transmutation system. For example, prioritizing older assemblies may produce a different feed composition than prioritizing high-burnup or high-enrichment assemblies, even when the same total mass is supplied. The simulation records these differences and preserves the connection between the supplied material and the assemblies from which it originated.

The primary contribution of this thesis is a material-flow model rather than a new accelerator-driven reactor design or a full evaluation of whether the United States should deploy accelerator-driven transmutation. A decision of that kind would still require detailed neutronics, thermal-hydraulics, materials, safety, economic, transportation, safeguards, and repository analyses. What this work provides instead is a model that connects a realistic, assembly-resolved representation of the U.S. inventory to an ADS facility inside a dynamic fuel-cycle simulation.

This connection provides a foundation for broader policy and deployment studies. A future analysis could add site-specific transportation, separations capacity, construction schedules, facility siting, operating costs, or alternative transmutation technologies. Those questions cannot be fully grounded in the present inventory, however, unless the model first represents the assemblies that actually exist and the rules by which they become available. By preserving assembly-level differences, this work makes the assumptions behind a transmutation scenario more transparent and its results more traceable.

The broader motivation is to ground the discussion of transmutation in how a real national inventory could supply such facilities over time, rather than in what a single idealized facility can destroy on its own. This emphasis matters because the performance of a technology in isolation may differ from its performance as part of an operating fuel cycle. Through the integration of assembly-level inventory data and an ADS

representation in CYCLUS, this thesis provides one step toward evaluating that larger system.

The remainder of this thesis is organized as follows. Chapter 2 reviews UNF management in the United States, the CYCLUS framework, accelerator-driven systems, the EINSTEIN program, and prior transmutation fuel-cycle simulations. It concludes by identifying the research gap addressed in this work. Chapter 3 describes the design and implementation of the `us_inventory` and ADS archetypes, the preparation of their input data, and the simulation scenarios considered. Chapter 4 presents the simulation results and compares the effects of the alternative inventory-selection policies. Finally, Chapter 5 discusses the implications and limitations of the results and identifies directions for future development.

Chapter 2

Background and Literature Review

2.1 Used Nuclear Fuel in the United States

UNF, also commonly called spent nuclear fuel, is nuclear fuel that has been removed from a reactor because it can no longer efficiently support the desired operating cycle. Although the fuel is no longer used in the reactor, it remains thermally hot and highly radioactive because it contains radioactive fission products, plutonium, and minor actinides produced during irradiation [6]. The term “used” is particularly appropriate because the discharged material is not completely depleted: most of its uranium remains, and the fuel also contains newly generated fissile isotopes that could potentially be recovered and reused.

The radioactivity and heat generation of UNF decrease with time, but they do so over multiple timescales. Short-lived fission products dominate the radiation field and decay heat immediately following discharge. At longer times, plutonium and the minor actinides become increasingly important because many of these isotopes have half-lives ranging from hundreds to tens of thousands of years. Consequently, untreated commercial UNF may require isolation over timescales on the order of 10^5 years before its radiotoxicity approaches that of the natural uranium ore from which the fuel was originally produced [3]. The long duration of this hazard is a central challenge in nuclear waste management.

The internationally accepted approach for the permanent isolation of UNF and high-level radioactive waste is disposal in a deep geologic repository [7]. The United States, however, does not currently operate a permanent repository for commercial UNF. Until such a repository or another disposition pathway becomes available, reactor licensees must store the fuel at or near the sites where it was generated [2], [6]. Newly discharged fuel is first placed in water-filled spent-fuel pools, which provide both cooling and radiation shielding. After several years of cooling, assemblies may be transferred to sealed dry-storage systems at independent spent fuel storage installations.

Because a national disposal system has not yet been implemented[2], the geographic distribution of UNF largely reflects the historical deployment of the U.S. reactor fleet. Most commercial fuel remains at operating or decommissioned nuclear power plant sites, including locations at which the reactor itself has been permanently shut down. The result is a distributed inventory that must eventually be transported, consolidated, processed, or disposed of[1].

Approximately 91,000 MTU from commercial pressurized-water reactors and boiling-water reactors had been permanently discharged in the United States. This inventory grows by roughly 2,000 MTU per year as operating reactors discharge fuel [8], and the current reactor fleet is projected to generate on the

order of 140,000 MTU in total by 2060. The U.S. Department of Energy (DOE) managed an additional inventory of approximately 2,446 metric ton of heavy metal (MTHM), excluding most naval fuel. Unlike the comparatively standardized commercial light-water-reactor inventory, DOE-managed UNF includes fuel from defense-production reactors, research and test reactors, early demonstration reactors, and other specialized programs. Altogether, UNF and primary reprocessing wastes were distributed across more than 100 locations in 39 states [1].

Table 2.1 summarizes the scale and distribution of this inventory. Commercial fuel dominates the national inventory by mass, while DOE-managed material is considerably more diverse in enrichment, fuel form, cladding, geometry, and irradiation history.

Table 2.1: The U.S. inventory of used nuclear fuel and primary reprocessing waste was distributed among commercial, DOE, research, and legacy waste sites. Data from [1], [8].

Inventory category	Reported quantity
Commercial UNF at nuclear power plant and ISFSI sites	90,963 MTHM
Permanently discharged PWR fuel	135,645 assemblies; 59,013 MTHM
Permanently discharged BWR fuel	179,466 assemblies; 32,023 MTHM
Commercial dry-storage canisters or casks	3,862
DOE-managed UNF	approximately 2,446 MTHM ^a
UNF at university, government, and commercial research sites	approximately 1 MTHM
Commercial nuclear power plant and ISFSI sites storing UNF	73 sites
Sites with all nuclear power reactors permanently shut down	20 sites
Operating commercial nuclear power reactors	94 reactors
Permanently shut-down commercial nuclear power reactors	27 reactors
DOE sites with UNF storage or research reactors	6 sites
University research-reactor sites	20 sites
Other government research-reactor sites	4 sites
Commercial research and development sites with UNF	4 sites
Primary reprocessing-waste or high-level-waste sites	4 sites

^a The detailed DOE inventory total depends on the treatment of naval fuel and commercial-origin fuel transferred to DOE custody. The value shown here follows the inventory basis used in this work. Quantities are rounded and may therefore differ slightly among inventory summaries.

The commercial inventory has also changed in composition over the history of the U.S. nuclear power program. In the early decades of light-water-reactor operation, discharged fuel commonly had burnups below 20 GWd/MTU, and initial enrichments were often near 2 wt% ²³⁵U. Improvements in fuel design, reactor operation, and fuel-management strategies allowed utilities to extend cycle lengths and extract more energy from each assembly. By the twenty-first century, average discharge burnup had increased to approximately 50 GWd/MTU, while typical initial enrichment approached 4.5 wt% ²³⁵U [1].

Figure 2.1 shows the increase in annual average burnup and initial enrichment for U.S. light-water-reactor discharges between 1968 and 2022. Although pressurized-water-reactor and boiling-water-reactor fuel follow somewhat different histories, both exhibit the same broad trend toward higher enrichment and higher discharge burnup.

Higher burnup changes the isotopic composition of discharged fuel. During irradiation, neutron capture in ²³⁸U produces plutonium isotopes, while successive neutron-capture and decay reactions generate neptunium, americium, and curium. Extending irradiation therefore generally increases the production of plutonium and minor actinides within an assembly, although the exact inventory depends on enrichment, reactor type, neutron spectrum, power history, and cooling time. The modern UNF inventory is consequently not uniform:

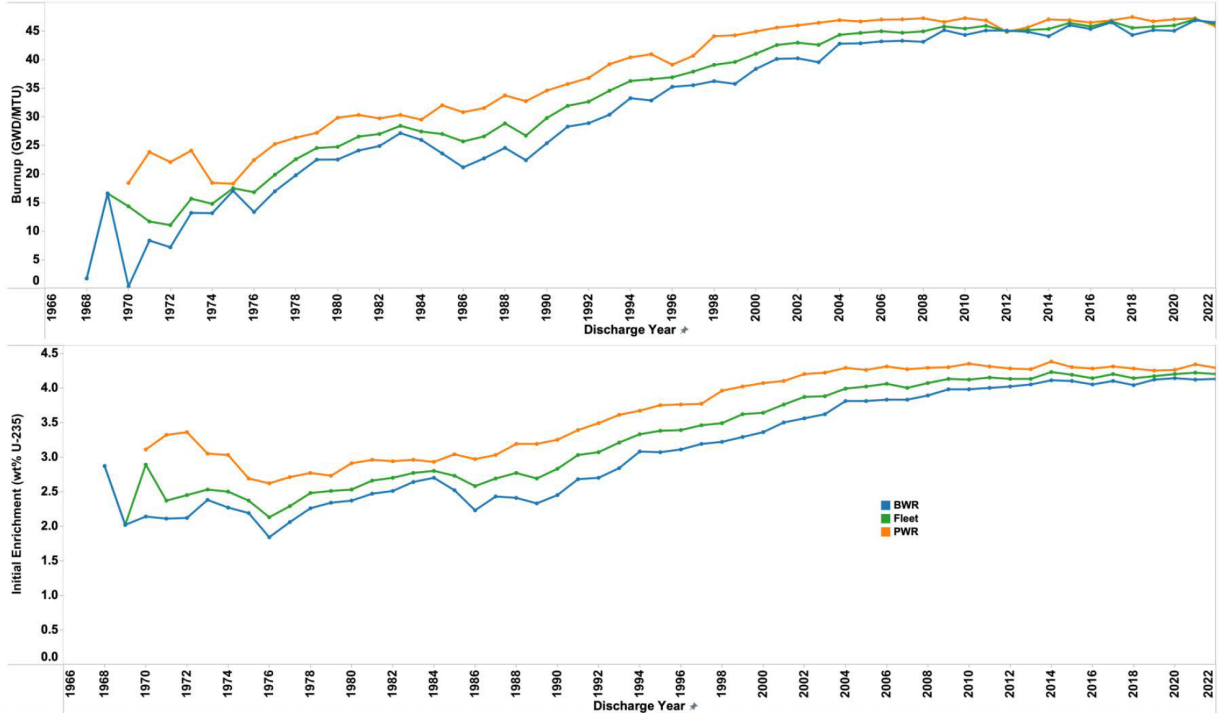


Figure 2.1: The annual average discharge burnup and initial ^{235}U enrichment of U.S. light-water-reactor fuel increased substantially between 1968 and 2022 [1].

assemblies discharged in different decades contain different quantities of fissile material, transuranic elements, fission products, and decay heat[1].

These compositional differences are important because the nuclides that control UNF behavior change with cooling time. Shortly after discharge, fission products dominate decay heat. In particular, ^{137}Cs and ^{90}Sr , together with their decay products, are major contributors over the first several decades. Their heat output decreases substantially as they decay. Over longer periods, transuranic isotopes—especially plutonium and americium isotopes—become the dominant contributors to the remaining heat and radiotoxicity [1].

This transition is significant for waste-management strategy. Storage can allow short-lived fission products to decay, but storage alone does not alter the fundamental long-term burden posed by plutonium and the minor actinides. These nuclides must otherwise remain isolated while they decay over much longer timescales. In contrast, actinides can be separated and irradiated in a suitable neutron spectrum, where fission converts them into shorter-lived fission products while releasing additional energy. Long-lived fission products may also be candidates for specialized transmutation, but the actinides are the principal component of UNF that can be destroyed through fission rather than managed solely by long-term storage and geologic isolation[9].

The increasing mass, geographic dispersion, and transuranic content of the U.S. UNF inventory therefore motivate the fuel-cycle simulations in this work. A realistic assessment of accelerator-driven transmutation must represent not only the total mass of available fuel, but also the differences among assemblies in discharge date, burnup, enrichment, location, and isotopic composition. These attributes determine which portions of the inventory can be selected, separated, and supplied to an accelerator-driven system.

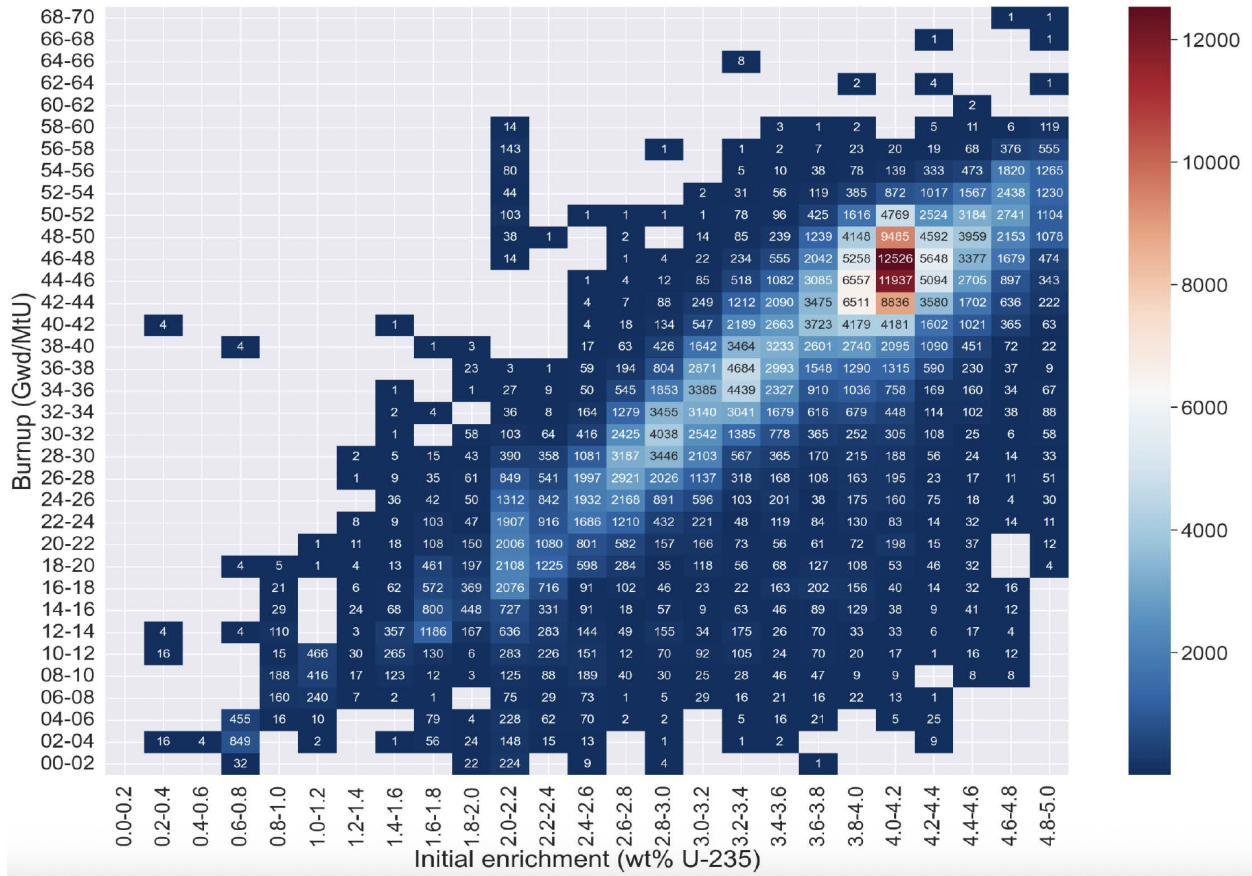


Figure 2.2: Burnup (GWd/MTHM) and initial enrichment (% ^{235}U) distributions for spent nuclear fuel assemblies in the Light Water Reactors (LWRs) inventory through December 2022 **carter'spent'2023**.

2.2 The Cyclus Fuel Cycle Simulation Framework

2.2.1 Framework Basics

CYCLUS is an open-source, agent-based nuclear fuel cycle simulation framework designed to represent the deployment, operation, and interaction of nuclear fuel cycle facilities over time. The framework tracks discrete facilities and material resources rather than representing the fuel cycle solely through aggregated fleet-level flows. This design supports analyses in which individual facility behavior, material availability, and institutional decisions influence system-level outcomes [10].

Agents in a CYCLUS simulation are organized into three hierarchical types: regions, institutions, and facilities. Regions form the highest organizational level and may represent countries, geographic areas, or other broad groupings. Institutions operate within regions and manage the deployment and retirement of facilities. Facilities represent physical or functional components of the nuclear fuel cycle, including mines, enrichment plants, reactors, separations facilities, storage facilities, and repositories [10].

A CYCLUS simulation advances through discrete time steps, commonly defined as months. During each time step, facilities may request resources, offer available resources, conduct material transactions, process their inventories, and update their internal states. Modeling facilities and time discretely allows the framework to capture effects such as asynchronous reactor refueling, temporary shortages of fuel-cycle materials, and differences in facility deployment and retirement schedules[11].

The behavior of each agent is defined by an *archetype*. Archetypes are modular software components that implement the physical behavior, operating logic, or decision rules of a region, institution, or facility. They are compiled independently from the CYCLUS kernel and loaded dynamically at the beginning of a simulation. This plug-in architecture allows alternative models of the same fuel-cycle process to be exchanged without modifying the core simulation framework [10], [12].

The CYCLUS kernel provides shared simulation services, including time management, agent deployment, resource tracking, data recording, and resource exchange. Archetypes access these services through the agent application programming interface. As illustrated in Figure 2.3, the kernel provides the central simulation environment, the agent interface connects the kernel to agent models, and independently developed archetype libraries provide the behavior of the simulated regions, institutions, and facilities.

2.2.2 Materials and Decay

Resources exchanged in CYCLUS are represented as discrete objects. The two primary resource types are *materials* and *products*. A material resource has an associated quantity and isotopic composition, whereas a product represents a non-nuclear resource whose quality may be described without a nuclide vector. Nuclear fuel cycle simulations therefore generally use material resources to represent fresh fuel, spent nuclear fuel, separated actinides, and radioactive waste [10].

A CYCLUS material object contains a total mass and a *Composition* object. The composition stores the relative abundance of the nuclides present in the material and is treated independently from the material quantity. This separation allows multiple material objects to reference the same composition and permits operations such as splitting, combining, and transmuting material while preserving mass accounting. Materials may also be stored in facility inventories and exchanged between agents through the resource exchange system.

Radioactive decay can be applied directly to material resources within CYCLUS. When a material is

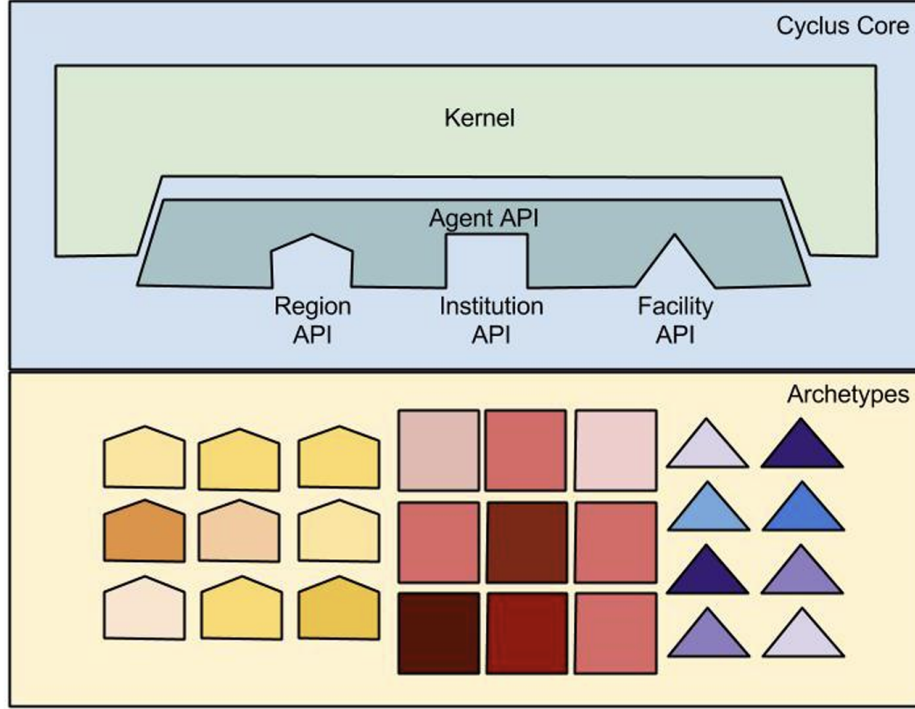


Figure 2.3: The CYCLUS simulation architecture separates the core kernel from dynamically loaded region, institution, and facility archetypes through the Agent application programming interface (API). Reproduced from [10].

decayed over a specified time interval, its nuclide vector is updated according to the radioactive decay chains available to the framework. The decay calculation uses nuclear data supplied through PyNE, including half-lives, decay constants, branching relationships, and daughter products. The total quantity of a material remains associated with the resource while its isotopic composition changes with time [10].

Decay in the simulations developed for this work is handled at the simulation level through the CYCLUS control block, with decay set to `lazy`. Under this setting the kernel decays a material’s composition on demand rather than at every time step, computing the decayed nuclide vector when a composition is next inspected and caching the result for reuse [10]. This choice retains the physical effect of radioactive decay over the multi-decade simulation horizon while avoiding the cost of decaying every inventory at every step. Decay is significant over this horizon: the roughly 14-year half-life of ^{241}Pu , for example, produces appreciable in-growth of ^{241}Am across a thirty-year run, altering the minor actinide vector that the inventory and accelerator-driven system models are designed to track.

2.2.3 The Dynamic Resource Exchange

Transactions among CYCLUS facilities are coordinated through the dynamic resource exchange (DRE). The DRE provides a general mechanism through which independently developed archetypes can exchange resources without requiring direct knowledge of one another’s internal implementation. Facilities communicate their demand and supply to the framework, and the exchange system determines a feasible set of transactions for each time step [10].

The exchange begins when consuming facilities generate requests for resources. A request identifies a

desired commodity, quantity, and, when relevant, a target material composition or other resource quality. Supplying facilities evaluate these requests and generate bids describing the resources they are able to provide. A bid links a particular offered resource to a request and may provide all or only part of the requested quantity[11].

Facilities may also assign preferences to possible trading partners or resources. Preferences allow a requester to rank otherwise feasible bids, for example by favoring a particular supplier, fuel composition, or material characteristic. In addition, suppliers and requesters may impose constraints such as production capacity, inventory availability, throughput, or minimum and maximum transaction quantities. These constraints define which combinations of trades are physically or operationally feasible.

The requests, bids, preferences, and constraints form a resource assignment problem. CYCLUS formulates this assignment as an optimization problem, solvable in either a linear or a mixed-integer form, and the DRE solves it to select a set of transactions that satisfies the applicable constraints while favoring higher-preference trades [11]. After the solution is determined, the participating facilities execute the selected transactions and update their inventories. This process is repeated at each simulation time step, allowing supply and demand relationships to change dynamically as facilities are deployed, retired, depleted, or otherwise alter their behavior.

By separating exchange coordination from facility-specific behavior, the DRE allows archetypes to be developed and combined modularly. A storage facility, reactor, separations plant, or repository can participate in the same exchange system as long as it implements the appropriate request, bidding, and trade interfaces. This separation is particularly important for this work because the inventory and accelerator-driven system models must interact with both existing and newly developed fuel cycle facilities.

2.2.4 Extensions Developed in This Work

This work extends the CYCLUS ecosystem and its `cycamore` archetype library [13] with two facility archetypes developed within the `einstein` archetype library. The `us_inventory` facility represents the U.S. spent nuclear fuel inventory at assembly resolution, reading isotopic compositions and assembly metadata from the STANDARDS 5.0.1 dataset[4], [5] and supplying material according to user-selected inventory selection policies. The `ads` facility represents an accelerator-driven subcritical burner, transforming a loaded core composition using a library of pre-computed Serpent depletion cases. Both archetypes participate in the resource exchange described above: `us_inventory` as a supplier whose selection policies determine which assemblies it offers against a given request, and `ads` as a requester whose feed preferences rank inventory and separations output. Their design, implementation, and the four simulation scenarios in which they are exercised are described in Chapter 3.

2.3 Accelerator-Driven Systems

ADSs combine a high-power particle accelerator with a subcritical nuclear assembly. They have been proposed as dedicated systems for the transmutation of plutonium, minor actinides, and selected long-lived fission products recovered from spent nuclear fuel [9]. The ADS concept is particularly relevant to this work because it provides a potential pathway for consuming minor-actinide-rich materials that present significant neutronic and safety challenges in critical reactors.

2.3.1 System Components and Operation

An ADS consists of three principal components: a high-energy proton accelerator, a spallation target, and a subcritical multiplying core. The accelerator produces a proton beam, often with an energy of several hundred megaelectronvolts to approximately one gigaelectronvolt. The beam strikes a heavy-metal target, where high-energy proton–nucleus interactions release neutrons through the spallation process. These source neutrons enter the surrounding subcritical core and induce fission reactions in the fuel [14], [15].

The target must generate a sufficiently intense neutron source while withstanding substantial beam heating, radiation damage, and mechanical loading. Heavy elements are attractive target materials because they produce relatively large numbers of neutrons when bombarded by high-energy protons. Proposed ADS targets include solid tungsten targets and liquid-metal targets based on lead or lead–bismuth eutectic. In some designs, a circulating liquid-metal target also contributes to heat removal [14], [15].

The core is designed to remain subcritical, such that

$$k_{\text{eff}} < 1, \quad (2.1)$$

where k_{eff} is the effective neutron multiplication factor. Unlike a critical reactor, the neutron population in an ADS core cannot sustain itself indefinitely without the external spallation source. Source neutrons initiate fission reactions and are multiplied within the subcritical core, allowing the thermal power produced by fission to exceed the power delivered by the proton beam [15], [16].

The accelerator therefore provides an additional mechanism for controlling the fission power. When the beam is interrupted, the external source of spallation neutrons is removed and the source-driven fission power decreases rapidly. This behavior is sometimes described by stating that the beam acts as an on–off switch for the fission chain. However, turning off the beam does not eliminate radioactive decay heat. The fuel and structures continue to generate heat after shutdown and require reliable residual heat removal [16].

Figure 2.4 illustrates the relationship among the accelerator, spallation target, and subcritical core.

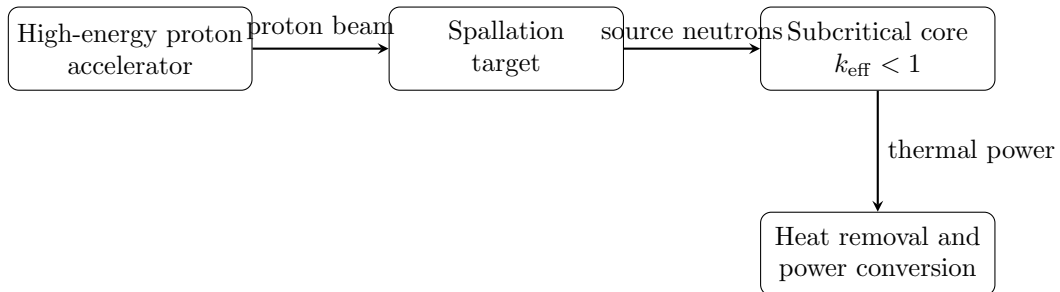


Figure 2.4: Principal components and energy flows of an accelerator-driven subcritical system. This original schematic is based on the ADS descriptions presented in [14], [15].

Accelerator-driven transmutation of used nuclear fuel received sustained attention in the United States during the late 1990s and early 2000s, most prominently through the Accelerator Transmutation of Waste (ATW) program, which developed system scenarios and integration roadmaps for dedicated subcritical burners [17]. Interest in the approach has since re-emerged in the context of managing the growing national inventory.

2.3.2 Minor-Actinide Transmutation

The principal motivation for using ADS technology for nuclear waste management is its ability to accommodate fuels containing large fractions of minor actinides. The minor actinides most commonly considered for transmutation are neptunium, americium, and curium. These nuclides contribute to the long-term radiotoxicity and decay heat of spent nuclear fuel. Their separation and irradiation in a dedicated transmutation system could reduce the quantity of long-lived transuranic material requiring disposal [9].

Fast neutron spectra are generally preferred for minor-actinide transmutation because fission becomes more competitive with radiative capture for many transuranic isotopes at higher neutron energies. A fast-spectrum system can therefore destroy minor actinides through fission rather than merely converting them into heavier actinides [9].

However, loading a critical fast reactor with a high concentration of minor actinides introduces important safety and control challenges. Minor-actinide-rich fuel generally has a smaller effective delayed-neutron fraction than conventional uranium- or plutonium-bearing fuel. Although delayed neutrons represent only a small fraction of the total fission neutron population, they substantially slow the response of a critical reactor to reactivity changes. A reduction in the effective delayed-neutron fraction therefore causes the reactor power to respond more rapidly to disturbances [16].

Minor-actinide loading may also reduce the magnitude of negative Doppler reactivity feedback. In conventional reactor fuel, increasing fuel temperature broadens neutron-absorption resonances and generally introduces negative reactivity. This prompt feedback opposes the temperature increase. The magnitude of this effect can be weaker in minor-actinide-rich, fast-spectrum fuels because of their isotopic composition and the reduced importance of resonance absorption in a fast spectrum [9], [16].

The combination of a reduced delayed-neutron fraction and weaker negative Doppler feedback makes a critical core with a high minor-actinide content more difficult to control. Critical fast-reactor concepts may therefore limit the amount of minor actinides loaded in the core or distribute them among fuel containing larger quantities of uranium or plutonium [16].

Subcritical operation provides an alternative approach. Because the core remains below criticality and relies on an external neutron source, an ADS can accommodate fuel compositions that would be difficult to use at high concentrations in a critical reactor. Subcriticality provides additional margin against prompt criticality and separates, to some extent, the control of the neutron source from the inherent reactivity of the fuel [16].

Subcriticality does not eliminate the need to evaluate reactivity accidents, cooling failures, material behavior, or decay heat removal. Nevertheless, it relaxes an important neutronic constraint on the amount of minor actinides that can be loaded into a dedicated transmutation system. This capability is the central reason that ADS technology is relevant to this thesis. The purpose of the modeled fuel cycle is to connect separated material from the U.S. spent nuclear fuel inventory with systems capable of consuming minor-actinide-rich streams that may not be suitable for concentrated use in critical reactors.

2.3.3 Target and Coolant Options

ADS concepts differ in their target materials, coolants, fuel forms, neutron spectra, and power levels. Lead and lead-bismuth eutectic are among the most frequently proposed heavy liquid metals because their large atomic masses support spallation neutron production and their high boiling temperatures permit low-pressure primary systems [14], [15].

Lead–bismuth eutectic has a substantially lower melting temperature than pure lead, reducing some of the operational challenges associated with coolant freezing. The most advanced current example is Multi-purpose Hybrid Research Reactor for High-tech Applications (MYRRHA), a lead–bismuth-cooled subcritical system under development at the Belgian Nuclear Research Centre. Its first phase, the MYRRHA Injector and low-energy accelerator (MINERVA) proton-accelerator and target facility, began construction in 2024 [18], making MYRRHA the closest existing demonstration of the accelerator-driven concept. However, neutron activation of bismuth produces polonium-210, which creates radiological and maintenance concerns. Both lead and lead–bismuth can also cause corrosion or erosion of structural materials. Their use therefore requires careful control of coolant chemistry, oxygen concentration, temperature, and flow conditions [15].

Pure lead avoids the production of polonium associated with bismuth but has a higher melting temperature. This increases the importance of freeze prevention during startup, shutdown, and off-normal operation. Despite this challenge, lead has been proposed both as a coolant and as a carrier medium for mobile-fuel ADS concepts [15], [19].

The ADS concept that provides the initial technical basis for the EINSTEIN project was developed to transmute material derived from the U.S. spent nuclear fuel inventory. The proposed system uses a high-energy proton accelerator, a spallation neutron source, and a lead-based subcritical system. The fuel concept contains plutonium and minor actinides in a mobile lead-based medium, permitting continuous or periodic processing of the fuel inventory [19].

Gas-cooled ADS concepts have also been investigated. Helium is chemically inert and avoids many of the corrosion and activation issues associated with heavy liquid metals. However, its low density and volumetric heat capacity require high-pressure operation and significant pumping power. The coolant and target selection therefore influences neutron production, material compatibility, plant efficiency, operating pressure, safety behavior, and the design of the primary heat-removal system [15].

2.3.4 Technical and Economic Limitations

Despite their potential advantages for transmutation, ADS facilities face substantial technical and economic challenges. Accelerator availability is one of the most important engineering concerns. A sudden beam interruption causes a rapid decrease in fission power and temperature. Restoration of the beam causes the power and temperature to increase again. Repeated beam interruptions can therefore impose cyclic thermal loads on the fuel, spallation target, coolant systems, and structural components [20].

The Organisation for Economic Co-operation and Development (OECD) Nuclear Energy Agency beam-interruption benchmark specifically examined the temperature transients caused by beam trips in a lead–bismuth-cooled, subcritical system. The study identified thermal fatigue of structural components as an important consequence of repeated beam interruptions [20]. More recent analysis of the China Initiative Accelerator Driven System has also evaluated fuel cladding fatigue under repeated beam-trip transients [21].

Producing a continuous, high-current proton beam with the availability required for a power-producing nuclear system is more demanding than operating an accelerator intended for intermittent research use. Fault-tolerant accelerator designs, redundant components, and rapid recovery procedures may reduce the number and duration of beam interruptions, but these features add complexity and cost [15].

The spallation target presents additional engineering challenges. It must tolerate concentrated beam heating, radiation damage, gas production, thermal stresses, and interactions with the coolant. Target replacement, inspection, and maintenance must also be performed in a highly radioactive environment [15].

ADS deployment additionally requires supporting fuel-cycle infrastructure. Spent nuclear fuel must first

be processed to recover the actinides intended for transmutation. The resulting minor-actinide-rich material must then be fabricated into fuel or introduced into an appropriate mobile-fuel medium. After irradiation, the material may require repeated separation and recycle to achieve substantial actinide destruction. These activities introduce material losses, secondary waste streams, safeguards requirements, worker exposure, and additional facilities[9].

An ADS also combines the capital cost and regulatory requirements of a nuclear reactor facility with those of a high-power accelerator. Furthermore, the accelerator consumes a portion of the electricity generated by the system. Comparative OECD/Nuclear Energy Agency (NEA) studies have therefore emphasized that the potential transmutation benefits of ADS technology must be evaluated alongside its fuel-cycle requirements, technical complexity, research and development needs, and economic performance[16].

These limitations make a system-level fuel-cycle analysis necessary. The performance of an ADS cannot be judged solely from the rate of transmutation inside its core. A complete assessment must also consider the availability and composition of feed material, separations capacity, facility deployment, accelerator and reactor throughput, recycle requirements, and the production of secondary waste streams. These interactions motivate the CYCLUS simulation models developed in this work.

2.4 The EINSTEIN Program

The EINSTEIN project is part of the ARPA-E NEWTON program. The NEWTON program was established to support technologies that enable the transmutation of used nuclear fuel and reduce the burden placed on permanent waste-disposal facilities [3]. The EINSTEIN project is led by the University of Illinois Urbana-Champaign [22].

NEWTON addresses the transmutation problem through three complementary technical categories. Category A focuses on technologies for generating and accelerating particle beams capable of initiating transmutation reactions. Category B focuses on the design, fabrication, and processing of targets or fuel forms containing the nuclides to be transmuted. Category C integrates the technologies developed in Categories A and B through system-level modeling, life-cycle analysis, techno-economic analysis, and databases of materials and components [3], [23]. UIUC is a Category C and therefore does not develop a single accelerator or transmutation target in isolation. Instead, it provides a common framework for comparing candidate systems and identifying the technical and economic conditions under which they could contribute to management of the U.S. used nuclear fuel inventory.

Within the EINSTEIN project, the term XDS refers collectively to subcritical transmutation systems sustained by an external neutron source. Within this framing, ADSs use an accelerator and spallation target to produce the source neutrons, while fusion-driven systems (FDSs) use neutrons generated by a fusion source. The two classes may differ substantially in neutron energy, source geometry, target and coolant materials, reliability constraints, and technology readiness. Nevertheless, both couple an external neutron source to a subcritical multiplying region and can therefore be represented within a common fuel-cycle and assessment framework. The initial EINSTEIN analyses include a high-energy, proton-driven, lead-cooled ADS and a 14.1 MeV fusion-neutron-driven, salt-cooled system [24].

The central objective of EINSTEIN is to develop a data-informed framework for rapidly assessing accelerator- or fusion-driven system (XDS) technologies according to both their transmutation performance and their financial implications. The project connects nuclear data, neutronics, materials compatibility, fuel-cycle behavior, and economic performance rather than evaluating each component independently. The

official project description identifies the development of a comprehensive life-cycle analysis framework spanning the full transmutation process, from the composition of the incoming used-fuel stream through irradiation, processing, and the resulting waste or product streams [22]. Experimental and computational activities are used to reduce uncertainties in neutron transport, material properties, and component performance.

The UIUC project team is organized around several interacting technical areas. Nuclear-data and diagnostics work identifies reactions whose uncertainties significantly affect neutron multiplication and transmutation predictions and develops experimental strategies to reduce those uncertainties. Neutronics analysis provides source-term, criticality, depletion, and transmutation calculations for selected XDS concepts. Materials research evaluates corrosion, irradiation damage, thermophysical properties, and the chemical behavior of lead coolants and molten salts. Experimental work investigates the stability and solubility of representative actinide surrogates and the compatibility of structural materials with the candidate coolant and fuel media [24].

These activities supply the inputs required for the fuel-cycle and techno-economic assessment. The fuel-cycle portion of the project represents the U.S. used nuclear fuel inventory, separations facilities, XDS facilities, recycle pathways, storage facilities, and waste streams in CYCLUS. The inventory characterization begins with national spent-fuel records and age-corrected isotopic compositions, which are used to define realistic feed streams for the modeled XDS technologies. New CYCLUS archetypes are then used to evaluate how the choice of feed material, facility throughput, deployment schedule, separations capacity, and material-selection policy affects system-level transmutation performance [24].

The techno-economic analysis uses the outputs of the fuel-cycle and component models to examine capital costs, operating costs, potential revenues, return on investment, and the sensitivity of economic conclusions to uncertain design parameters. This coupling is important because a design that performs well in a stand-alone neutronics calculation may still require an impractical number of facilities, excessive separations capacity, costly transportation, or unavailable supporting infrastructure. Conversely, a concept with a lower per-facility transmutation rate may be more attractive if it is easier to deploy, operate, and integrate with the existing nuclear fuel cycle.

The work presented in this thesis lies primarily within the fuel-cycle modeling portion of EINSTEIN. It develops the `us_inventory` and `ads` archetypes in the `einstein` library and uses them to connect the characteristics of the national used-fuel inventory with candidate XDS deployment scenarios. The resulting simulations provide material-flow, facility-scaling, and inventory-depletion information that can be combined with the neutronics, materials, and economic analyses performed elsewhere in the project. In this way, the thesis contributes the system-level link between the physical design of an XDS and its potential effect on the U.S. used nuclear fuel inventory.

2.5 Prior Fuel Cycle Simulations of Transmutation Systems

Research on accelerator-driven transmutation has progressed through several distinct stages. Early studies established the physics of coupling an intense accelerator-produced neutron source to a subcritical multiplying medium. Subsequent programs developed and tested individual enabling technologies, including high-current accelerators, liquid-metal spallation targets, subcriticality-monitoring methods, minor-actinide fuels, and lead or lead-bismuth coolant systems. More recent studies have examined how dedicated transmutation facilities could be incorporated into regional or national nuclear fuel cycles.

These bodies of work do not all address the same modeling question. Reactor physics studies generally

evaluate neutron multiplication, power distributions, safety coefficients, and nuclide destruction within one core. Technology demonstrators test particular components or operating procedures. Fuel-cycle scenario studies instead examine the deployment of many reactors and supporting facilities over several decades, tracking quantities such as plutonium and minor-actinide inventories, separations requirements, fuel fabrication capacity, and waste production. The present work belongs primarily to the third category, but its facility models are informed by the first two.

2.5.1 European Demonstrator Programs

Europe has sustained the most continuous multinational program of ADS research. Its development pathway has included conceptual system studies, critical and subcritical experiments, a megawatt-class liquid-metal spallation-target experiment, and the staged development of the MYRRHA research facility. These programs have demonstrated important elements of an ADS, although no European facility has yet operated as a high-power, minor-actinide-burning ADS.

One influential early European concept was Carlo Rubbia’s Energy Amplifier. The Energy Amplifier combined a high-energy proton accelerator, a heavy-metal spallation target, and a subcritical fast-spectrum blanket. Early versions emphasized energy production from thorium, while later analyses also considered waste transmutation and fast-spectrum operation [25]. The importance of this work was not that it produced a directly deployable fuel-cycle design, but that it articulated a complete accelerator–target–blanket system and stimulated coordinated European research on spallation targets, lead-based coolants, subcritical reactor physics, and partitioning and transmutation.

European research was subsequently coordinated through programs including EUROTRANS, which investigated an experimental ADS and the European Facility for Industrial Transmutation European Facility for Industrial Transmutation (EFIT). EFIT was conceived as a lead-cooled, fast-spectrum minor-actinide burner rather than as a general-purpose power reactor. Its design studies addressed accelerator requirements, target integration, subcritical-core design, dedicated minor-actinide fuel, and the supporting closed fuel cycle. In the “double-strata” strategy commonly associated with these studies, conventional critical reactors would manage most uranium and plutonium recycling, while a smaller fleet of dedicated ADS facilities would receive and destroy concentrated minor-actinides[26].

The EFIT studies are particularly relevant to fuel-cycle analysis because they treated an ADS not as an isolated core but as one component of a broader fleet. Regional scenario calculations examined the number of ADS facilities required to stabilize or reduce minor-actinide inventories and estimated the associated separations and fuel-fabrication capacities. These analyses demonstrated that transmutation performance depends not only on the destruction rate of one ADS, but also on cooling delays, reprocessing losses, fuel-fabrication throughput, deployment dates, and the availability of separated feed material [26].

The principal European engineering demonstrator is MYRRHA developed by Belgian Nuclear Research Centre (SCK CEN) in Belgium. MYRRHA is designed as a lead–bismuth-eutectic-cooled, fast-spectrum research system capable of subcritical accelerator-driven operation and, in the broader design concept, critical operation. Its purposes include demonstrating ADS technology, conducting fuels and materials irradiations, supporting radioisotope production, and providing experimental data relevant to minor-actinide transmutation [14], [18].

MYRRHA is being implemented in phases. The first phase, MINERVA, comprises the lower-energy portion of the proton accelerator and associated target stations. Construction of this phase began in 2024. The later phases would extend the accelerator and couple it to the subcritical reactor. MYRRHA therefore

represents the most advanced European program directed toward a power-scalable ADS, but it should not yet be described as an operating accelerator-driven reactor [18].

Several predecessor experiments were developed to reduce the technical risk of MYRRHA. The Generator of Uninterrupted Intense Neutrons at the Lead VENUS Reactor (GUINEVERE) project coupled the Générateur de Neutrons Pulsés Intenses, third generation with continuous beam (GENEPI-3C) deuteron accelerator to the fast-spectrum configuration of the Vulcan Experimental Nuclear Study reactor (VENUS-F) fast lead critical and subcritical assembly at SCK CEN. GUINEVERE was designed principally to investigate reactor-physics questions, including the determination and online monitoring of subcriticality, source-jerk and beam-interruption methods, core loading procedures, and the transition between critical and subcritical configurations [27], [28], [29]. It was a low-power experimental facility rather than a fuel-burning demonstrator, but it provided evidence that an accelerator could be coupled to a fast subcritical assembly and that the system’s reactivity could be measured using realistic source-operation modes.

A separate enabling-technology experiment was the MEGAwatt Pilot Experiment, or MEGAwatt Pilot Experiment (MEGAPIE), operated at the Swiss spallation neutron source Swiss Spallation Neutron Source (SINQ) in 2006. MEGAPIE exposed a circulating lead–bismuth target to a continuous 590 MeV proton beam at approximately 0.77 MW. It was the first liquid-metal spallation target operated in the megawatt beam regime and supplied experimental data on neutron production, heat removal, liquid-metal operation, activation products, structural behavior, and post-irradiation examination [30], [31], [32]. MEGAPIE did not include a multiplying reactor core, so it did not demonstrate an integrated ADS. Its importance was instead the validation of one of the most demanding ADS components under an irradiation environment approaching that required for a full-scale system.

France contributed to these European programs through Commissariat à l’Énergie Atomique et aux Énergies Alternatives (CEA), National Centre for Scientific Research (CNRS), and other institutions, as well as through national fuel-cycle scenario studies. CEA developed the Commellini-Sicard (COSI) simulation code to model reactor fleets and associated fuel-cycle facilities over long time horizons. COSI studies examined transitions from light-water reactors to sodium-cooled fast reactors and compared homogeneous, heterogeneous, and dedicated minor-actinide transmutation strategies [33]. Other French and European studies compared ADS-based transmutation with minor-actinide burning in critical fast reactors, including differences in fuel-cycle cost, inventory reduction, and supporting facility requirements [34], [35].

Taken together, the European programs established a progression from component experiments to integrated demonstrator design and fleet-level scenario analysis. GUINEVERE addressed source coupling and subcriticality monitoring, MEGAPIE addressed a high-power liquid-metal target, EFIT provided a reference industrial transmuter design, and MYRRHA seeks to integrate many of these capabilities into a research facility. Their scenario studies also showed that the viability of ADS deployment cannot be inferred from single-core transmutation efficiency alone.

2.5.2 United States Transmutation Programs, 1990s–2000s

Modern U.S. accelerator-driven transmutation work grew partly from studies at Los Alamos National Laboratory in the late 1980s and early 1990s. Bowman et al. proposed coupling an intense accelerator-produced neutron source to a subcritical multiplying and transmuting medium. The concept considered the destruction of transuranic elements and selected long-lived fission products, together with energy production from fertile material [36]. This work helped establish the technical foundation for the (ATW) program.

ATW was developed during the 1990s as a system for treating commercial spent nuclear fuel using

partitioning, accelerator-driven transmutation, and repeated recycle. A reference ATW complex included front-end separations, minor-actinide and plutonium fuel preparation, one or more accelerator-driven subcritical burners, recycle separations, and waste conditioning. The U.S. Department of Energy submitted a development roadmap to Congress in 1999, identifying required research in accelerators, spallation targets, separations, fuels, materials, reactor physics, safety, and systems analysis [17], [37].

The roadmap treated ATW explicitly as a fuel-cycle system rather than merely as a reactor. This was important because repeated recycle is required to achieve high actinide destruction. No irradiation cycle destroys every atom fed to the system: some actinides remain in the discharged fuel, some are converted into other actinides, and small fractions are lost during chemical processing and fabrication. Consequently, the overall reduction in waste burden depends on the efficiency and capacity of the complete recycle loop.

The follow-on Advanced Accelerator Applications (AAA) program broadened the scope of the U.S. effort to include waste transmutation, energy applications, and enabling accelerator technologies. Program planning documents considered the reliability of high-power proton accelerators, liquid-metal and solid targets, subcritical blanket configurations, separations based on aqueous and pyrochemical methods, and fabrication of fuels containing high fractions of transuranic elements. Much of this work remained at the level of design studies, laboratory research, and component development. No integrated U.S. ATW or AAA engineering demonstrator was constructed.

Systems studies performed for ATW and related programs compared combinations of light-water reactors, critical fast reactors, and accelerator-driven systems. These studies investigated questions such as whether plutonium should first be recycled in light-water reactors, whether ADS facilities should consume all transuranics or only minor actinides, and how many recycle passes would be required. Their results repeatedly showed that conclusions were sensitive to separations efficiency, fuel cooling time, accelerator availability, support ratio, and the assumed growth or decline of nuclear power [17], [37].

After the large ATW and AAA programs ended, U.S. ADS work continued through more focused reactor-physics, materials, target, and systems studies. A recent and particularly relevant example is the Argonne National Laboratory concept reported by Gohar, Cao, and Kraus. Their design was developed specifically to consume plutonium and minor actinides derived from the U.S. spent nuclear fuel inventory. It employs a 1 GeV proton beam, a lead-based spallation target, and a subcritical fast blanket with mobile fuel containing plutonium and minor actinides [19].

Gohar et al. used Monte Carlo transport and depletion calculations to examine several blanket compositions and estimate the actinide consumption achievable over decades of full-power operation. The study concluded that a small fleet of large ADS units could consume the minor-actinide inventory assumed in its analysis. However, its scope was a set of reference ADS designs and their depletion performance. It did not dynamically model the assembly-level U.S. inventory, site-to-site material movement, separations deployment, facility-construction schedules, or competition among facilities for particular spent-fuel resources. Thus, it provides the principal reactor design basis for the ADS model in this thesis, but not the system-level model needed to determine how that design would be supplied and deployed.

2.5.3 Ongoing Construction: CiADS

China began a national ADS development program through the Chinese Academy of Sciences in 2011. Early work included the (China Lead-based Reactor (CLEAR)) program and a series of lead-bismuth experimental loops, zero-power facilities, and non-nuclear test systems. These activities supported the development of coolant technology, structural materials, fuel assemblies, components, and operating methods for a lead-cooled

subcritical system [38].

The China initiative Accelerator Driven System (CiADS), led by the Institute of Modern Physics of the Chinese Academy of Sciences, is now under construction in Huizhou. The integrated facility is designed to couple a superconducting proton linear accelerator, a lead–bismuth spallation target, and a lead–bismuth-cooled subcritical reactor. Published design studies describe a first-stage reactor with a maximum thermal power of approximately 10 MW. The initial core uses oxide fuel and is intended primarily to demonstrate the engineering feasibility, control, safety, and coupled operation of the ADS rather than to serve immediately as an industrial-scale minor-actinide burner [39], [40].

CiADS is significant because it moves beyond isolated component tests toward construction of an integrated accelerator–target–subcritical-reactor facility. Its research program addresses accelerator reliability, beam transport, target behavior, lead–bismuth technology, reactor control, thermal hydraulics, and protection systems. In the proposed operating strategy, core power is controlled through the proton-beam power because the subcritical core does not contain a conventional critical-reactor control system for routine power regulation [39], [41].

Dynamic and transient studies of CiADS have analyzed beam trips, beam overpower, loss of flow, loss of heat sink, and station-blackout conditions. These calculations indicate substantial thermal margins for the low-power demonstrator while identifying longer-term concerns such as lead–bismuth corrosion, creep, and fatigue under repeated thermal cycling [40]. The project therefore provides an important test of whether the theoretically attractive subcritical characteristics of ADSs can be realized in a maintainable engineering system.

CiADS should not be interpreted as a completed demonstration of a commercial transmutation fuel cycle. The first-stage system is much smaller than the gigawatt-thermal designs commonly assumed in industrial fuel-cycle studies, and its initial mission emphasizes technology demonstration. Nevertheless, operational experience from CiADS could reduce major uncertainties presently treated parametrically in systems models, including beam availability, component lifetime, maintenance frequency, achievable capacity factor, and the coupling between accelerator and reactor operation.

2.5.4 Fuel-Cycle Simulation Studies

The studies described above span a wide range of spatial and temporal resolutions. Core-design calculations may represent individual fuel pins and detailed nuclide vectors but usually consider only one reactor and a limited number of irradiation cycles. Fuel-cycle simulators generally model decades of fleet evolution, but they aggregate fuel into batches, reactor types, or material streams. This trade-off between reactor detail and system breadth is central to understanding the gap addressed by this thesis.

The OECD/NEA regional-scenarios study extended this type of analysis to groups of countries with different nuclear policies and fuel-cycle infrastructure. It used the EFIT design as the reference ADS and calculated the number of transmuters and associated fuel-cycle facilities required to stabilize minor-actinide inventories across a regional system [26]. This work showed that regional sharing of specialized facilities could reduce duplication of expensive separations and fabrication plants. However, its material representation was at the level of national or regional inventory streams rather than individual discharged assemblies.

France’s COSI code has been used extensively for dynamic scenario analysis. COSI represents reactor fleets, fuel fabrication, irradiation, cooling, reprocessing, storage, and waste production over time. Applications have included transitions from the existing French light-water-reactor fleet to sodium-cooled fast reactors, along with comparisons of minor-actinide management options [33]. These studies are highly detailed in their

treatment of fleet evolution and industrial fuel-cycle capacity, but they generally use reactor and fuel batches characterized by representative compositions rather than a geospatially explicit assembly inventory.

European projects also compared the economic and material consequences of minor-actinide transmutation in critical fast reactors and ADS facilities. Chabert et al. compared no-transmutation, fast-reactor, and ADS options, while Merino-Rodríguez et al. cross-checked economic and mass-balance calculations for scenarios involving dedicated ADS burners [34], [35]. These studies demonstrated that ADS-based strategies may achieve concentrated minor-actinide destruction but incur additional costs associated with the accelerator, dedicated fuel fabrication, and supporting recycle facilities.

The Belgian Analysis of Nuclear-fuel-cycles Including Chain and Cost Assessment (ANICCA) code provides a more recent example directly related to ADS deployment. ANICCA is a dynamic nuclear-fuel-cycle simulator developed to study Belgian fleet evolution and waste management. A published verification and case study examined the effect of adding an ADS to a Belgian nuclear phase-out scenario. The model tracked fleet-level materials and showed the potential reduction of minor-actinide inventories, but it did not model individual commercial fuel assemblies or source-site-specific selection [42].

Other studies have focused on detailed depletion within an ADS rather than deployment of the surrounding fuel cycle. These calculations optimize core geometry, fuel loading, neutron spectrum, or recycle strategy and report nuclide-specific transmutation rates. They are essential for supplying performance parameters to a system model, but they do not themselves answer how quickly suitable material becomes available, where it is stored, which assemblies are selected, or how infrastructure limits deployment [19].

Across the literature reviewed here, system-level ADS studies generally use one of three material representations. The first is an equilibrium mass-flow model, in which annual production and destruction rates are balanced without explicitly simulating each facility. The second is a dynamic fleet model in which reactor types and fuel-cycle plants are deployed over time and exchange aggregated batches. The third is a detailed core or depletion model in which a single ADS is simulated at high isotopic resolution but the external fuel cycle is prescribed rather than dynamically represented.

These approaches have provided estimates of ADS support ratios, required fleet size, equilibrium actinide inventories, separations capacity, fuel-cycle cost, and repository benefits. They have not generally represented the existing U.S. used-fuel inventory as a set of discrete assemblies carrying site, discharge date, burnup, enrichment, mass, and composition metadata. In the literature reviewed for this thesis, no prior ADS fuel-cycle study was identified that dynamically selected individual assemblies from the site-resolved U.S. inventory and supplied them through separations to parameterized ADS facilities.

That distinction is important for the U.S. case. The national inventory is not an equilibrium material stream produced by a uniform reactor fleet. It is a heterogeneous legacy inventory accumulated over several decades and stored at geographically dispersed operating and shutdown sites. Assemblies differ in age, burnup, enrichment, reactor type, remaining heavy-metal mass, isotopic composition, and storage location. A model that begins with a representative average composition cannot evaluate how alternative inventory-selection policies change the material supplied to separations and transmutation facilities.

The present work addresses this gap by combining an assembly-resolved U.S. inventory facility with dynamically deployed ADS and supporting fuel-cycle facilities in CYCLUS. It does not replace detailed neutronics, depletion, materials, or economic models. Instead, it provides the system layer through which those models can be connected to the actual characteristics and availability of U.S. used nuclear fuel. This representation permits analysis of inventory-selection policies, throughput limits, facility scaling, deployment timing, material availability, and the evolution of the remaining national inventory.

Table 2.2: Prior accelerator-driven transmutation programs and studies addressed different levels of technology and fuel-cycle resolution. Information summarized from [18], [19], [26], [27], [31], [40], [42].

Program or study	Type	Principal scope	Material resolution
GUINEVERE	Experimental facility	Accelerator coupling, fast subcritical physics, and online reactivity monitoring	Core configurations; no fuel-cycle deployment
MEGAPIE	Component experiment	Megawatt-class liquid lead–bismuth spallation target	Target and neutron-source behavior only
MYRRHA	Staged demonstrator	Accelerator, target, fast-spectrum research reactor, fuels, materials, and ADS operation	Core and irradiation experiments; supporting scenario studies
EFIT / European scenarios	Reference design and fleet studies	Dedicated minor-actinide burning and regional deployment	Aggregated fuel batches and regional inventories
ATW / AAA	U.S. program and system studies	Separations, recycle, accelerator, target, and subcritical burner	Process streams and representative fuels
Gohar et al.	ADS design and depletion study	Lead-based mobile-fuel ADS for the U.S. actinide inventory	Prescribed blanket compositions
CiADS	Integrated demonstrator under construction	Coupled accelerator, spallation target, and low-power subcritical reactor	Detailed core and component models
COSI / AN-ICCA studies	Dynamic fuel-cycle simulation	Fleet transitions, facility capacities, inventories, and waste	Reactor fleets and aggregated fuel batches
This work	Dynamic fuel-cycle simulation	U.S. inventory selection, separations, ADS deployment, throughput, and inventory evolution	Individual assemblies with associated meta-data

2.6 Research Gap

The literature reviewed in Section 2.5 demonstrates substantial progress in accelerator-driven transmutation, but the principal areas of progress have remained separated. Reactor-physics and engineering studies have developed increasingly detailed representations of accelerators, spallation targets, subcritical cores, fuels, coolants, and transmutation performance. Fuel-cycle studies have examined the deployment of reactors, separations plants, fabrication facilities, and waste-management infrastructure over long periods. National spent-fuel databases have also developed sufficiently detailed records to characterize individual discharged fuel assemblies. However, these capabilities have not generally been combined in a single simulation. In the literature reviewed for this thesis, no published study was identified that dynamically couples a real, assembly-resolved U.S. spent nuclear fuel inventory to an ADS facility inside a fuel-cycle simulator and evaluates how the selection of particular assemblies affects transmutation-system deployment and the evolution of the remaining national inventory.

European ADS research has produced the most mature combination of experimental facilities, component demonstrations, core calculations, and regional fuel-cycle studies. GUINEVERE investigated accelerator coupling and subcriticality monitoring, MEGAPIE demonstrated a liquid-metal spallation target under a megawatt-class proton beam, and MYRRHA has advanced the engineering design and staged implementation of an integrated ADS research facility [14], [18], [27], [28], [31]. European fuel-cycle tools have also examined fleet transitions and minor-actinide transmutation. For example, COSI has been used to evaluate French reactor and fuel-cycle transitions, while ANICCA has modeled the introduction of an ADS into a Belgian nuclear phase-out scenario [33], [42]. These studies provide detailed information about European reactor fleets, fuel-cycle infrastructure, and transmutation strategies, but their feed materials are represented as reactor-specific or scenario-specific fuel streams and batches. They do not begin with the discrete assembly records of the existing U.S. inventory or evaluate alternative rules for selecting particular U.S. assemblies for processing.

The U.S. Accelerator Transmutation of Waste and Advanced Accelerator Applications programs addressed many policy-relevant aspects of a complete transmutation system. Their roadmaps and system studies considered separations, repeated actinide recycle, fuel fabrication, accelerator reliability, spallation-target design, subcritical burners, and final waste management [17], [37]. These efforts recognized that the performance of an ADS could not be assessed independently from the facilities that prepare, recycle, and manage its fuel. Nevertheless, the principal ATW and AAA studies were conducted before the development of current agent-based fuel-cycle simulators and before the availability of the present assembly-level STANDARDS database. Consequently, their national inventory was represented through assumed aggregate material streams, reference isotopic compositions, and projected production rates rather than through the individual assemblies that constitute the existing U.S. inventory.

More recent U.S. work has improved the physical basis of the ADS itself. Gohar, Cao, and Kraus developed a lead-based ADS concept specifically for consuming plutonium and minor actinides derived from U.S. spent nuclear fuel and used detailed transport and depletion calculations to estimate actinide consumption over the operating life of the system [19]. This study provides the initial ADS design basis used in the present work, but its treatment of the national inventory begins with the total assumed masses of plutonium and minor actinides. It does not model the discharged assemblies from which those materials would be recovered, the order in which assemblies would be selected, or the time-dependent interaction among inventory, separations, and ADS facilities.

CiADS addresses a different part of the problem. Its published analyses focus on the engineering behavior

of an integrated accelerator, spallation target, and low-power subcritical reactor. Studies have examined proton-beam control, reactor transients, beam trips, thermal margins, coolant behavior, and structural fatigue [21], [38], [39], [40]. This work is essential for demonstrating whether ADS technology can be operated reliably, but CiADS is an engineering demonstrator rather than a simulation of national fuel-cycle deployment. Its analyses therefore do not address how a heterogeneous legacy inventory would be selected, transported, separated, and allocated among multiple future transmutation facilities.

Existing fuel-cycle and inventory studies come closest to connecting these areas. The STANDARDS 5.0.1 Database contains detailed information about the spent nuclear fuel (SNF) inventory in the U.S. with full isotopic composition. Peterson et al. described the planned use of Unified Database (UDB) information in system models such as Next-Generation System Analysis Model (NGSAM) and in fuel-cycle transition analyses using ORION (ORION) [5]. Bae et al. subsequently trained a neural network on UDB depletion data and implemented the resulting model in CYCLUS, enabling efficient prediction of spent-fuel compositions for reactor batches with varying burnup and enrichment [43]. These studies established that realistic U.S. data can improve fuel-cycle simulation. The former described broader integration of the database with system and fuel-cycle tools, while the latter reproduced UDB-derived isotopic behavior without exposing or dynamically transacting the underlying private assembly records. Neither work connected an active assembly inventory to an ADS burner or examined how the choice among available assemblies changes the feed supplied to that burner.

The gap addressed in this thesis is therefore narrower than the absence of either detailed ADS models or detailed inventory data. The missing capability is their integration at the system level. This work represents individual U.S. spent-fuel assemblies as selectable resources within an agent-based fuel-cycle simulation. Assembly records derived from STANDARDS 5.0.1 retain attributes relevant to material selection, including available heavy-metal mass, discharge date, burnup, initial enrichment, and isotopic composition [1]. The `us_inventory` archetype supplies these resources through the CYCLUS Dynamic Resource Exchange rather than first collapsing the national inventory into a single average recipe.

The inventory model is coupled to an ADS archetype whose material transformation is based on precomputed Serpent depletion results. The simulation can therefore evaluate multiple inventory-selection policies, including selection by availability, age, burnup, and enrichment, while maintaining material accounting as assemblies are supplied and their available masses are depleted. The results presented in Chapters 3–4 quantify how these policies change the mass and characteristics of material supplied to the modeled fuel-cycle facilities and how they alter the composition of the inventory remaining after transmutation.

This integration does not by itself constitute a complete recommendation for U.S. waste policy. Questions involving facility siting, transportation approval, safeguards, separations readiness, cost, financing, and repository performance require additional models and institutional assumptions. Rather, the contribution of this thesis is to provide a U.S.-specific material-flow foundation on which those analyses can be built. By connecting an assembly-resolved legacy inventory to an ADS representation in CYCLUS, the work enables transmutation scenarios to be evaluated using the actual heterogeneity and availability of U.S. spent nuclear fuel instead of an assumed national-average feed.

Chapter 3

Methodology

3.1 Overview of the Simulation Approach

The simulations in this work are performed with `CYCLUS`, the agent-based nuclear fuel cycle simulator introduced in Chapter 2. Two facility archetypes were developed within the `einstein` archetype library: `us_inventory`, which represents the United States inventory of UNF at the resolution of individual assemblies, and `ads`, which represents an accelerator-driven subcritical burner. These are combined with facilities from the `CYCAMORE` library—`Separations` to partition discharged fuel into product and waste streams, `Mixer` to blend separated streams into a burner feed, and `Sink` to receive final waste—to form a complete transmutation fuel cycle, shown in Figure 3.1.

Figure 3.1: Fuel cycle flowsheet used in this work. The `us_inventory` facility supplies discharged assemblies to a `Separations` facility; separated material is irradiated in the `ads` facility, and the resulting streams are recycled or sent to waste.

The remainder of this chapter describes each archetype in turn—`us_inventory` in Section 3.2 and `ads` in Section 3.3—and then defines the specific scenarios built from them in Section 3.4.

All scenarios share a common set of simulation-level settings. Time advances in monthly steps of 30.4375 d, beginning in January 2025, over a horizon of either 30 years (360 steps) or 100 years (1200 steps). Radioactive decay is enabled at the simulation level through the `CYCLUS` control block with decay set to `lazy`, so that material compositions are decayed on demand over the multi-decade horizon rather than at every time step [10]. The full set of runs and their per-scenario parameters is given in Section 3.4.

3.2 The `us_inventory` Archetype

The `us_inventory` archetype represents the existing U.S. UNF inventory as an active material supplier. Rather than collapsing the national inventory into a single resource with a prescribed mass and representative composition, it preserves individual assembly records, because the U.S. inventory is heterogeneous and the order in which assemblies are selected changes the mass and isotopic composition delivered to downstream facilities.

Each assembly is stored as a separate inventory bin with its own mass, operating history, storage information, composition, and geographic provenance. During a simulation the facility answers material

requests through the CYCLUS DRE (Chapter 2) and selects an eligible assembly according to a user-specified policy, reducing that assembly’s mass and the total inventory as material is supplied. It inherits from `cyclus::Facility` and `cyclus::toolkit::CommodityProducer`, so it acts as both a facility agent and a producer of a configured output commodity. The facility performs no scheduled action each timestep; all of its behavior is triggered either when the input data are loaded at initialization or when it participates in resource exchange.

3.2.1 Purpose and Assembly-Level Data Source

The source data are assembly-level records derived from STANDARDS 5.0.1, which associates individual U.S. spent-fuel assemblies with facility, storage, irradiation-history, and calculated isotopic information. This detail distinguishes assemblies that would otherwise be merged into a facility- or national-average composition [5], [44].

For the simulations in this thesis, the STANDARDS data were provided as a directory of 75 line-delimited files, one per represented facility. The file count is not hard-coded: the `data_dir` parameter names a directory, and every file whose name ends in `.txt` or contains `.jsonl` is read, so the archetype can run on a subset of the database without recompilation.

Each nonempty line is one assembly record, converted into assembly metadata and a nuclide mass map. Because the export wraps each dictionary-like record in an outer JSON string, a purpose-built parser removes that wrapper and accepts both single- and double-quoted strings. It reads the facility name, storage-location name, assembly identifier, storage type, discharge year, initial enrichment, maximum burnup, initial uranium mass, and a dictionary of nuclide masses in grams. Unrecognized fields are ignored, and records with no nonzero composition are skipped.

The nuclide names are converted to the integer ZZAAAM identifiers used by CYCLUS and assembled into a `cyclus::Composition`; the total assembly mass is converted from grams to kilograms and stored with the record. Each assembly is held in a `Bin` structure, summarized in Table 3.1.

Table 3.1: Fields stored in each `Bin` structure.

Field	Description
<code>assembly_id</code>	Unique internal key for the assembly.
<code>orig_id</code>	Original database assembly identifier.
Facility, location	Source facility and storage-location names.
Storage type	Wet or dry storage.
Available mass	Current material mass, in kilograms.
Discharge year	Year the assembly was discharged.
Maximum burnup	Reported maximum burnup.
Initial enrichment	Initial uranium enrichment.
Initial uranium mass	Initial mass of uranium.
Derived fractions	Fissile, plutonium, and minor-actinide mass fractions.
Coordinates	Latitude and longitude of the source facility.
Composition	Complete CYCLUS composition of the assembly.

Because the original identifier is not unique across the dataset, the archetype builds an internal key from the facility name, storage location, original identifier, and load position, stored in `assembly_id`; the raw value is kept in `orig_id`. Each bin’s available mass starts as the sum of its reported nuclide masses, and a running total tracks the material remaining in the facility.

Three composition fractions used by the selection policies are computed once at load time. For assembly i with total mass m_i and nuclide mass $m_{X,i}$ for nuclide X , the fissile mass fraction is

$$f_{\text{fissile},i} = \frac{m_{^{233}\text{U},i} + m_{^{235}\text{U},i} + m_{^{239}\text{Pu},i} + m_{^{241}\text{Pu},i}}{m_i}, \quad (3.1)$$

where the numerator sums the masses of ^{233}U , ^{235}U , ^{239}Pu , and ^{241}Pu in assembly i , and m_i is its total mass. The total plutonium mass fraction is

$$f_{\text{Pu},i} = \frac{\sum_a m_{a\text{Pu},i}}{m_i}, \quad (3.2)$$

where a runs over all plutonium isotopes present in assembly i . The minor-actinide mass fraction is

$$f_{\text{MA},i} = \frac{\sum_a m_{a\text{Np},i} + \sum_a m_{a\text{Am},i} + \sum_a m_{a\text{Cm},i}}{m_i}, \quad (3.3)$$

where a runs over all isotopes of neptunium, americium, and curium in assembly i . These fractions are precomputed so the selection policies need not traverse the full nuclide composition on every request.

The required inputs are the output commodity, `outcommod`, and the assembly-data directory, `data_dir`. An optional maximum throughput, `throughput_kg`, in kilograms per time step, is effectively unlimited by default but can be reduced to represent a constraint on inventory retrieval, transportation, or processing.

3.2.2 Assembly Provenance and Geographic Coordinates

Preserving assembly provenance connects inventory-selection results to later transportation, siting, and infrastructure analyses. The STANDARDS records identify each assembly's facility and storage location but not its coordinates, so an optional SQLite database associates a facility name with latitude and longitude.

The database path is supplied through `coordinates_file`. If it is empty, the archetype still runs but every bin keeps the default pair $(0, 0)$. When a database is provided, it is opened read-only and queried for the facility name, taking an exact match where one exists and otherwise a prefix match. All assemblies from a facility share one coordinate pair, cached after the first lookup. If no entry matches, the archetype warns and leaves that facility at $(0, 0)$.

Every completed transfer is recorded in a custom CYCLUS output table, `us_inventorySupply`, with one row per supplied trade (Table 3.2). The transferred material separately carries the composition built from the selected bin, so a transaction can be traced to both its source assembly and its material composition.

3.2.3 Assembly-Selection Policies

The archetype's principal methodological feature is its ability to choose among individual assemblies by explicit ranking policies. In resource exchange the solver decides which offers and requests match, but the supplier still chooses which physical resource fulfills an accepted trade; `us_inventory` uses that supplier-side choice to model different assumptions about access to the legacy inventory.

The policy is set through `selection_policy` and checked at initialization, where an unrecognized value stops the simulation with a descriptive input error. For each trade the archetype scans the eligible bins and keeps the one ranked highest under the active criterion, skipping empty bins and bins without a valid composition. Table 3.3 lists the 15 policies.

Table 3.2: Columns written to the `us_inventorySupply` output table.

Column	Description
Agent ID	Identifier of the supplying agent.
Time	Simulation time step of the transfer.
Facility	Source facility name.
Storage location	Source storage-location name.
Storage type	Wet or dry storage.
Assembly ID	Original assembly identifier.
Initial uranium	Initial uranium mass of the assembly.
Latitude, longitude	Coordinates of the source facility.
Quantity	Mass supplied in the trade.
Selection policy	Policy in force for the simulation.

Table 3.3: Assembly-selection policies implemented by the `us_inventory` archetype. Each policy ranks the eligible assemblies before a trade is fulfilled.

Policy	Assembly selected
<code>first</code>	First eligible bin in load order; a baseline that does not rank by a physical property.
<code>older</code>	Smallest discharge year.
<code>newer</code>	Largest discharge year.
<code>highest_burnup</code>	Largest reported maximum burnup.
<code>lowest_burnup</code>	Smallest reported maximum burnup.
<code>highest_enrichment</code>	Largest initial enrichment.
<code>lowest_enrichment</code>	Smallest initial enrichment.
<code>highest_initial_uranium</code>	Largest initial uranium mass.
<code>lowest_initial_uranium</code>	Smallest initial uranium mass.
<code>highest_fissile</code>	Largest fissile fraction (Eq. 3.1).
<code>lowest_fissile</code>	Smallest fissile fraction.
<code>highest_plutonium</code>	Largest plutonium fraction (Eq. 3.2).
<code>lowest_plutonium</code>	Smallest plutonium fraction.
<code>highest_minor_actinide</code>	Largest minor-actinide fraction (Eq. 3.3).
<code>lowest_minor_actinide</code>	Smallest minor-actinide fraction.

Ranking keeps the first eligible bin as the initial candidate and replaces it only when a later bin is strictly better under the criterion, so ties keep their load-order priority.

Two optional inputs bias the pool toward a particular source. `preferred_facility` names a specific facility, and `storage_type` restricts the pool to `wet` or `dry` storage. These filters are preferential rather than exclusive: the matching assemblies are searched first, and if they cannot supply a trade the rest of the inventory is searched, with a warning the first time this fallback occurs. This keeps the archetype consistent with the DRE, which expects a bidding facility to remain able to supply an accepted trade.

Whether a request may be partially filled is governed by `allow_partial`. Selection first requires that a bin cover the full requested mass; only if none can, and `allow_partial` is true, is a smaller quantity permitted. Each accepted trade is drawn from a single bin—the archetype does not combine undersized assemblies—which keeps a one-to-one relationship between each trade response and the assembly recorded in `us.inventorySupply`.

3.2.4 Material Bidding, Trading, and Accounting

Bidding and trading are handled in two stages: the facility first advertises how much material it can offer, and then, once trades are awarded, withdraws a specific assembly for each one.

When there are requests on the output commodity, the facility bids. Because the source filters are preferential, the whole remaining inventory is eligible, so the facility can fall back to nonmatching assemblies. The compositions of the eligible bins are combined into a single mass-weighted bid composition,

$$\mathbf{x}_{\text{bid}} = \frac{\sum_{i \in \mathcal{E}} m_i \mathbf{x}_i}{\sum_{i \in \mathcal{E}} m_i}, \quad (3.4)$$

where $\mathcal{E}$ is the set of eligible bins, m_i is the remaining mass of bin i , and $\mathbf{x}_i$ is its vector of nuclide mass fractions. The maximum mass offered is

$$M_{\text{offer}} = \min(M_{\text{eligible}}, M_{\text{throughput}}), \quad (3.5)$$

where $M_{\text{eligible}} = \sum_{i \in \mathcal{E}} m_i$ is the total mass of the eligible pool and $M_{\text{throughput}}$ is the configured per-time-step throughput. This value enters the portfolio as a capacity constraint. With partial fulfillment disabled, the archetype does not bid on a request larger than it can offer.

The bid composition represents the available pool but does not replace policy-based selection of the source assembly. After the solver awards trades, each one is matched to a source bin by the procedure in Section 3.2.3, so the material returned is built from the selected assembly’s composition, not the averaged bid composition.

Each accepted trade is limited by the requested quantity, the selected bin’s remaining mass, and the throughput remaining in the step, and fulfillment stops once the throughput or the inventory is exhausted. After each response the archetype updates

$$m_{i,t+1} = m_{i,t} - m_{\text{supplied}}, \quad (3.6)$$

$$M_{\text{inv},t+1} = M_{\text{inv},t} - m_{\text{supplied}}, \quad (3.7)$$

$$R_{t+1} = R_t - m_{\text{supplied}}, \quad (3.8)$$

where $m_{i,t}$ is the available mass of the selected assembly i , $M_{\text{inv},t}$ is the total mass remaining in the facility, R_t is the throughput remaining in the current step, and m_{supplied} is the mass delivered by the trade.

The per-assembly remaining mass is also persisted as archetype state, one value per loaded assembly. Restarting from a checkpoint reloads the assembly metadata and compositions from the source files, overwrites their initial masses with the persisted values, and rebuilds the inventory total from them. The simulation is stopped if the number of persisted values does not match the number of loaded bins, since that indicates the data directory changed after the checkpoint.

After each successful transaction the archetype appends a row to `us_inventorySupply`, giving a time-dependent history of which assemblies were selected, where they originated, how much they supplied, and which policy produced that sequence—the basis for the policy comparisons in Chapter 4.

3.3 The ads Archetype

The archetype represents an ADS operating as a transuranic burner. It receives an actinide-bearing feed from upstream facilities, accumulates one core loading, irradiates it for a prescribed cycle, and discharges the surviving actinides together with the fission products generated during the cycle. Like `us_inventory` (Section 3.2), it inherits from `cyclus::Facility` and `cyclus::toolkit::CommodityProducer` and so is both a facility agent and a producer of an output commodity; unlike `us_inventory`, it is time-dependent, advancing an irradiation cycle each timestep and participating in the DRE on both the request and the bid side.

3.3.1 Reduced-Order Transmutation Model

Solving neutron transport and depletion inside a multi-decade fuel-cycle simulation, for every cycle of every burner, is computationally intractable. The `ads` archetype therefore uses a *reduced-order* model: the physics is computed once, offline, by high-fidelity depletion calculations, and at runtime the archetype performs only a case lookup and an interpolation. This mirrors the recipe-based treatment of irradiation established in CYCLUS [10].

The offline physics is a library of SERPENT depletion cases (Section 3.3.2), each recording the time evolution of the full nuclide inventory of a fixed loading irradiated at constant power. The library is loaded once when the agent enters the simulation and held in memory.

Buffers and cycle. The facility holds three material buffers: `feed` accumulates received material, `core` holds the loading under irradiation, and `product` holds discharged material awaiting trade. The feed and core buffers share the `capacity` parameter—the mass of one core—while the product buffer is unbounded. Operation is a two-state batch cycle keyed on whether the core is occupied. An empty core is refueling; once `refuel_time` has elapsed and the feed holds a full core, one `capacity` of feed is loaded. An occupied core irradiates until its cycle counter reaches `cycle_time`, at which point it is discharged. The composition is not

altered during irradiation—the whole transformation is applied as a single transition at discharge—because the core does not participate in resource exchange while loaded. Two persisted counters track progress through the irradiation and refueling periods so the operating state survives a simulation snapshot.

Feed requests. Feed may be requested on several input commodities, listed in `incommods` with optional per-commodity preferences. The archetype requests the space available in its feed buffer, and the per-commodity requests are mutually exclusive alternatives in one portfolio, so it draws one core’s worth from whichever supplier the exchange prefers rather than the full amount from each. An optional `feed_recipe` may be attached; when it is empty, the request is for blank material and the archetype accepts whatever composition the supplier offers. This is the mode used here, since the feed composition varies with the inventory-selection policy and with any upstream separations.

Simulation time to irradiation time. CYCLUS advances in timesteps, whereas the SERPENT cases are tabulated in irradiation days. The parameter `days_per_step`, Δ_d , converts between them, so the irradiation time read at discharge is

$$t_{\text{irr}} = n_{\text{cycle}} \Delta_d, \quad (3.9)$$

where n_{cycle} is the cycle length in timesteps (`cycle_time`). If Δ_d is not set, it is derived from the timestep duration. This is what gives the cycle length physical meaning: a longer cycle addresses a later, more-depleted point on the trajectory.

Case selection. At discharge the core is matched to the library case that best represents it. The core is reduced to the mass fractions of three actinide groups, normalized to the total actinide mass:

$$f_{\text{MA}} = \frac{m_{\text{Np}} + m_{\text{Am}} + m_{\text{Cm}}}{m_{\text{act}}}, \quad f_{\text{Pu}} = \frac{m_{\text{Pu}}}{m_{\text{act}}}, \quad f_{\text{U}} = \frac{m_{\text{U}}}{m_{\text{act}}}, \quad (3.10)$$

where each element mass (m_{Np} , m_{Am} , m_{Cm} , m_{Pu} , m_{U}) is summed over all isotopes of that element, $m_{\text{act}} = \sum_{Z \geq 89} m$ is the total actinide mass, and the minor actinides are neptunium, americium, and curium. Nuclides below $Z = 89$ are excluded, so fission products in recycled feed do not affect selection. The same fractions are computed for each case from its initial ($t = 0$) actinide vector. Each case c is scored by

$$S_c = w_P \frac{|P - P_c|}{\max(P, P_c)} + \sum_{g \in \{\text{MA}, \text{Pu}, \text{U}\}} |f_g - f_{g,c}|, \quad (3.11)$$

where P and P_c are the facility and case powers, f_g and $f_{g,c}$ are the core and case group fractions, and $w_P = 10$ weights the power term. The lowest-scoring case wins; the large w_P makes the power tier dominate so a core is not matched across power levels on composition alone. A single nearest case is chosen—there is no averaging across cases—and interpolation happens only along that case’s time axis.

Composition transformation. The selected case gives nuclide masses $m_n(t_k)$ at tabulated times t_k . For the interval $[t_{\text{lo}}, t_{\text{hi}}]$ bracketing t_{irr} , each nuclide is linearly interpolated,

$$m_n(t_{\text{irr}}) = (1 - \alpha) m_n(t_{\text{lo}}) + \alpha m_n(t_{\text{hi}}), \quad \alpha = \frac{t_{\text{irr}} - t_{\text{lo}}}{t_{\text{hi}} - t_{\text{lo}}}, \quad (3.12)$$

over every actinide and fission product the case tracks, and the result becomes the discharged composition. Outside the tabulated range the composition is clamped to the nearest endpoint rather than extrapolated: depletion saturates as the most fissile nuclides are consumed, so a straight-line projection past the last point would overpredict burnup and eventually yield negative masses. A cycle that routinely runs past the range signals a misconfigured time mapping or a need for longer depletion runs, not a case for extrapolation.

Mass balance. The transformed composition holds both surviving actinides and the fission products made during irradiation, so mass is conserved to the extent SERPENT tracks it. With $M(t)$ the total tracked (actinide plus fission-product) mass of the case at time t , the discharged mass scales from the loading M_{in} by

$$R = \min\left(1, \frac{M(t_{\text{irr}})}{M(0)}\right), \quad M_{\text{out}} = R M_{\text{in}}, \quad (3.13)$$

the clamp at unity preventing a slightly-above-one ratio from creating mass. The residual $(1 - R) M_{\text{in}}$ leaves the simulation as a recorded mass defect, representing mass SERPENT does not carry in the discharged tables—chiefly the mass-energy released in fission, together with any light or untracked nuclides. Writing $a_c(t) = M_{\text{act},c}(t)/M_c(t)$ for the actinide fraction of the matched case, the destroyed actinide mass reported for the cycle is

$$\Delta M_{\text{act}} = M_{\text{in}} a_c(0) - M_{\text{out}} a_c(t_{\text{irr}}). \quad (3.14)$$

This is a summary quantity only; the discharged isotopics come directly from Equation 3.12. Discharged material is then offered on the output commodity under a capacity constraint equal to the product mass, and accepted trades draw from the product buffer for routing to storage, separations, or disposal.

3.3.2 Serpent Depletion Cases

These depletion cases were produced at UIUC, and this section documents the model they represent so that the archetype’s inputs can be interpreted. Each case is a continuous-energy Serpent 2 Monte Carlo depletion calculation [CITE: leppanenStatusSerpentMonte2025] that tracks the full nuclide inventory (`set inventory all`), producing the time-dependent actinide and fission-product mass tables the archetype later reads; the material volumes required for depletion normalization are obtained by a Monte Carlo volume calculation.

The model is an axisymmetric cylindrical ADS core built from concentric z -cylinders, summarized in Table 3.4. A central liquid-lead spallation target sits on the core midplane, surrounded by a thin structural target wall and, above it, an evacuated beam duct through which the proton beam reaches the target. The fuel occupies the annular region around the target tube, and the whole assembly is enclosed in structural material and an outer container. A vacuum (black) boundary condition is applied at the outermost surface, so neutrons leaking past the container are lost. The single target, single homogeneous fuel region, and absence of assembly- or pin-level detail are consistent with the reduced-order role of these cases: they are intended to characterize the bulk transmutation behavior of a core-worth of feed, not its detailed spatial power distribution.

Table 3.4: Principal regions of the Serpent ADS model. Dimensions are taken directly from the input decks; all cylinders are coaxial about the z -axis.

Region	Radius (cm)	Axial span (cm)	Material
Spallation target	0–17.5	150–210	Liquid lead
Beam duct (vacuum)	0–17.5	210–400	Void
Target wall	17.5–18.5	149–400	Structural
Fuel	18.5–150	0–300	Minor-actinide feed
Container / structure	150–250	–100–400	Structural

Fuel Compositions and the Case Set

The cases differ in the composition loaded into the fuel region, which is what makes them a *library* rather than a single run. Each fuel is a transuranic feed dispersed in a lead matrix. Two representative cases illustrate the range spanned by the library. In one, the feed is essentially pure minor actinide dispersed in lead; in another, the feed is an equal split of plutonium and minor actinide (a “50/50” Pu–MA blend) in lead. These two feeds bracket the transmutation regimes of interest: a minor-actinide-only charge maximizes the mass of minor actinide destroyed per cycle but is the most reactive to load, whereas adding plutonium softens the loading while shifting some of the destruction burden onto bred and fissioned plutonium. The initial actinide mass of each case, taken from the day-zero row of its actinide-mass table, is the quantity the archetype’s **capacity** parameter should reproduce, so that a loaded core matches the case it is compared against.

Each case is depleted along a common “Day [d]” time axis. The irradiation history is built up from a sequence of chained restart runs: an initial segment resolves the first fraction of a day with fine steps (for example, cumulative times of 0.001, 0.1, 0.2, and 0.5 d) to capture the rapid early transient in short-lived fission products, and each subsequent segment reads the binary restart file written by the previous one and advances the total burnup further (for example, from 730 to 760 d). Reassembled, these segments give the continuous nuclide inventory versus irradiation time that the archetype consumes: an actinide-mass table (`plot_data_15_actinide_masses_full.txt` or `plot_data_06_actinide_masses.txt`), a fission-product-mass table (`plot_data_14_fission_product_masses.txt`) on the same axis, and a power history (`plot_data_03_power_vs_time.txt`) used for case matching. The archetype reads a case’s composition at the irradiation time implied by the cycle length by linearly interpolating these tables along the day axis, so the resolution of the discharged composition is set by the density of the Serpent time grid rather than by the CYCLUS time step.

3.3.3 Irradiation-Event Recording

Besides the standard CYCLUS tables, the archetype writes a custom **AdsEvents** table so each cycle can be reconstructed without parsing the full isotopic history. Every row carries the agent id, time, event type, matched case, and a mass value (Table 3.5); the matched case is blank on a load, since matching happens at discharge.

The three discharge rows are written only when positive. The **ActinideDestroyed** series is the primary performance metric, and its cumulative sum gives the actinide mass a facility removes from the fuel cycle. With the standard CYCLUS **Transactions** and **ExplicitInventory** tables, these events close the actinide balance: the actinide initially present equals the sum destroyed, held in buffers, and resident downstream.

Table 3.5: Event types recorded in the `AdsEvents` output table.

Event	Recorded value
Load	Mass loaded from feed into the core.
Discharge	Core mass entering the transformation, M_{in} , before Equation 3.13; the matched case is recorded here.
ActinideDestroyed	Destroyed actinide mass, Equation 3.14.
FissionProducts	Fission-product mass in the discharged material.
MassDefect	Mass not retained, $(1 - R) M_{\text{in}}$.

3.4 Simulation Scenarios

The two archetypes developed in this chapter are exercised in a set of fuel-cycle simulations that share one flowsheet and differ along three axes: the composition burned in the ADS, the length of the simulated horizon, and whether the discharged material is recycled. Two burner configurations are used, corresponding to the two representative Serpent cases of Section 3.3.2: a minor-actinide-only core (Case 3) and a 50/50 plutonium–minor-actinide core (Case 4). Each configuration is run over a 30-year and a 100-year horizon, in a once-through mode and a recycling mode, for eight simulations in total.

3.4.1 Fuel Cycle Flow

Every simulation uses the same chain of facilities. The `us_inventory` facility supplies discharged assemblies on the `snf` commodity under the `highest_minor_actinide` selection policy (Section 3.2.3), so that the assemblies richest in neptunium, americium, and curium are drawn first. A `CYCAMORE Separations` facility partitions this material: the minor actinides are routed to the burner feed, while uranium, plutonium (except in Case 4), and the fission products fall to a leftover stream sent to a waste `Sink`. The ADS irradiates the separated feed and discharges the surviving actinides together with the fission products produced during the cycle. Depending on the scenario, that discharge is either sent to a `Sink` (once-through) or returned to `Separations` (recycling).

The two burner configurations shape the feed differently. For the minor-actinide-only core (Case 3), `Separations` extracts a single stream of neptunium, americium, and curium at 99% efficiency and feeds it directly to the ADS. For the 50/50 core (Case 4), the feed cannot be taken straight from separations: the U.S. inventory holds roughly ten times more plutonium than minor actinide, so a stream drawn from it is far from an equal split. `Separations` therefore produces two streams—one of minor actinides and one of plutonium—and a `CYCAMORE Mixer` blends them in a fixed 0.5/0.5 mass ratio to reproduce the Case 4 Serpent feed. The minor-actinide stream is the limiting one, so surplus plutonium accumulates in its buffer; this minor-actinide-limited feed rate is a physical consequence of the inventory composition rather than a modeling constraint.

The recycling and once-through modes differ only in the routing of the burner discharge. In once-through mode the discharge is a terminal waste stream held in a `Sink`. In recycling mode it is returned to `Separations`, which recovers the surviving minor actinides—and, for Case 4, plutonium—into fresh feed, so that actinides make repeated passes through the burner while uranium and fission products are removed to waste on each pass.

All simulations share the remaining settings. Time advances in monthly steps of 30.4375 d from a start of January 2025, radioactive decay is evaluated lazily over the horizon, and explicit inventory tracking is

enabled. The inventory, separations, and mixing facilities each operate at a throughput of 250,000 kg per step, large enough that material handling does not throttle the burner.

3.4.2 Simulation Runs

The two burner configurations fix the ADS and separations parameters summarized in Table 3.6. Each core mass matches the day-zero actinide mass of the corresponding Serpent case, so a loaded core maps onto the case it is compared against; the cycle lengths differ by more than the mass alone, as the large minor-actinide core is irradiated for ten years per cycle while the smaller 50/50 core is cycled every two years.

Table 3.6: ADS and separations configuration for the two burner cases. Cycle and refueling times are given in monthly time steps, with the equivalent duration in parentheses.

Parameter	Case 3 (MA only)	Case 4 (50/50 Pu–MA)
Initial core mass	27,000 kg	4,418 kg
Cycle length	120 steps (10 yr)	24 steps (2 yr)
Refueling time	6 steps (6 mo)	4 steps (4 mo)
Power level	3000 MW	3000 MW
Separated streams	Np, Am, Cm	Np, Am, Cm and Pu
Feed shaping	Fed directly	CYCAMORE Mixer, 0.5/0.5
Serpent case	892_vt_ma_Pb	1.3_vt_5050_puma_Pb

Each configuration is simulated over both horizons and in both modes, giving the eight runs listed in Table 3.7. Across most of these runs the once-through and recycling modes produce nearly identical histories, because the burner is cycle-limited rather than feed-limited: within the horizon it completes the same number of cores whether or not the discharge is recycled, and the recovered material only displaces inventory that was still available. The exception is the minor-actinide-only core over the 100-year horizon, where the large ten-year core draws down the accessible minor-actinide inventory and recycling sustains operation beyond what the once-through supply allows. The comparison between recycling and once-through in Chapter 4 therefore focuses on that pair of runs; for the remaining cases a single representative run is shown, since the two modes coincide.

Table 3.7: Simulation runs. Each burner configuration is run over both horizons and in both fuel-management modes.

Run	Fuel case	Mode	Horizon	Steps
1	MA only	Once-through	30 yr	360
2	MA only	Recycle	30 yr	360
3	MA only	Once-through	100 yr	1200
4	MA only	Recycle	100 yr	1200
5	50/50 Pu–MA	Once-through	30 yr	360
6	50/50 Pu–MA	Recycle	30 yr	360
7	50/50 Pu–MA	Once-through	100 yr	1200
8	50/50 Pu–MA	Recycle	100 yr	1200

Chapter 4

Results

4.1 Inventory Characterization

Before any transmutation results are presented, this section characterizes the inventory that the `us_inventory` archetype ingests, establishing both that the STANDARDS 5.0.1 dataset is loaded correctly and the scale and distribution of the material available to the fuel cycle. Because the inventory load is deterministic—the same directory of assembly records always produces the same pool—these results depend only on the input data and the parser, not on any simulation dynamics.

Reading the 75 assembly-record files yields 315,976 individual assemblies, totalling 8.69×10^4 t of heavy metal, drawn from 61 facilities for which geographic coordinates are available. Summing the minor-actinide content of every assembly, as defined in Equation 3.3, gives a total accessible minor-actinide inventory of 103.8 t. This minor actinide is only 0.12% of the total heavy metal, a consequence of the low neptunium, americium, and curium content of discharged light-water-reactor fuel. Put in terms of the burner, the entire national minor-actinide inventory amounts to fewer than four ADS cores at the 27 t core loading of the minor-actinide-only case (Table 3.6). This scarcity is the central constraint on the transmutation mission and recurs throughout the results: the burner is limited far more by the availability of minor actinide than by any reactor parameter.

Figure 4.1 shows the geographic distribution of this material, with each facility placed at its coordinates and shaded by the mass of spent fuel it supplies over a full-horizon run. Because the run draws down the accessible inventory, the drawn mass approaches the available inventory at each site, so the figure doubles as a map of where the national inventory resides. The distribution is strongly weighted toward the eastern and south-central United States, where the largest single sites each supply on the order of 2,500–2,700 t, while the smaller western sites contribute a few hundred tonnes. This spatial concentration is the starting point for the provenance analysis of Section 4.2, since any policy that draws assemblies in a particular order also implies a particular geographic sequence of retrieval.

Establishing this baseline confirms that the archetype ingests the assembly-level dataset faithfully and quantifies the material the transmutation fuel cycle has to work with. The remainder of this chapter examines how that material is selected (Section 4.2), how it is transmuted in the ADS (Section 4.3), and how the accessible inventory evolves under recycling (Section 4.4).

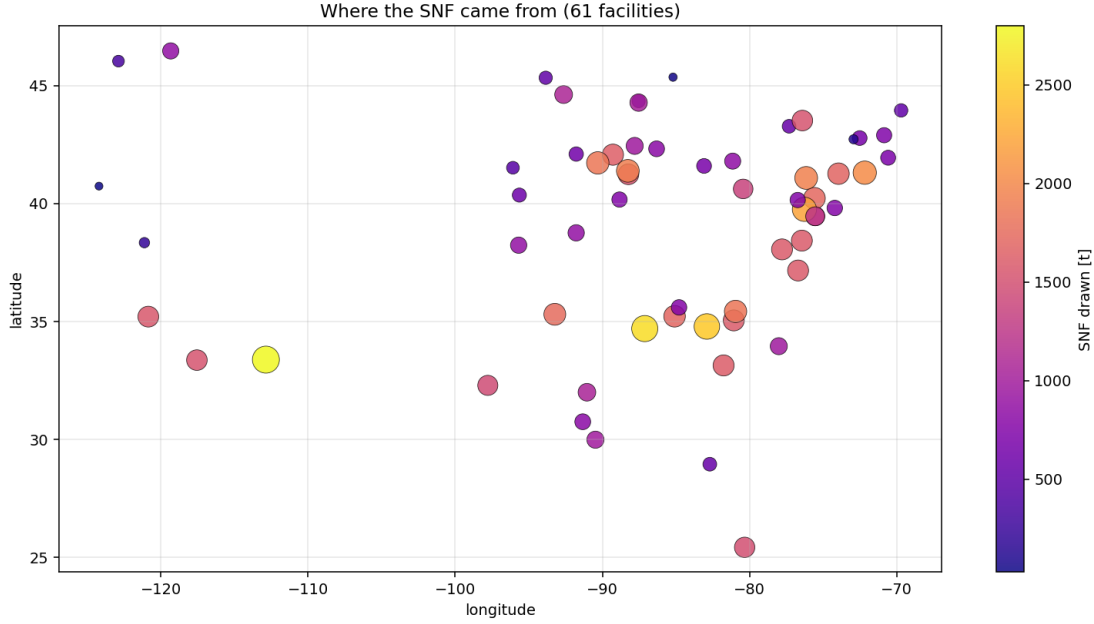


Figure 4.1: Geographic distribution of the spent nuclear fuel supplied by the `us_inventory` archetype. Each marker is one of the 61 facilities with known coordinates, positioned by latitude and longitude and shaded and sized by the mass of spent fuel drawn over the full simulation horizon. The inventory is concentrated in the eastern and south-central United States.

4.2 Effect of Selection Policy on the Feed

The `us_inventory` archetype selects individual assemblies by an explicit ranking policy (Section 3.2.3), and that choice determines the mass and composition of material delivered to separations and, ultimately, to the burner. The selection mechanism acts on the static inventory pool and is deterministic in the assembly ordering it produces: it does not depend on downstream reactor dynamics. Its effect on the feed can therefore be characterized directly, from the inventory data and from standalone `us_inventory` runs, independently of the full fuel cycle, and that is how it is presented in this section. The production scenarios of Chapter 4 all use `highest_minor_actinide` (Section 3.4.1); the comparisons here show what that choice buys relative to the alternatives.

4.2.1 Delivery Trajectory

Because every policy eventually draws from the same finite pool, the total minor actinide any policy can deliver is identical: the cumulative curves all terminate at the same 103.8 t (Section 4.1). What the policy changes is the *trajectory*—how quickly minor actinide accumulates as assemblies are drawn. Figure 4.2 shows the cumulative minor actinide delivered against the number of assemblies drawn, for the three representative policies of set A, with a horizontal line marking one 27 t minor-actinide-only core.

The separation is large. Ranking by `highest_minor_actinide` reaches a full core after roughly 4.8×10^4 assembly draws, whereas `lowest_minor_actinide` requires about 1.28×10^5 —a factor of 2.7 more material handled for the same core. The unranked `first` baseline falls between them, at about 7.7×10^4 draws, and its curve is visibly segmented: because `first` draws facility by facility in load order, its slope steps up and down as it moves through sites of differing minor-actinide quality, whereas the ranked policies produce smooth,

monotonically curving trajectories. The timing here is expressed in assemblies drawn rather than calendar time, since the mapping to time depends on the per-step retrieval throughput, which is not the subject of this comparison.

This is the practical face of the scarcity established in Section 4.1. With the national inventory holding fewer than four cores of minor actinide in total, the order of retrieval governs how much spent fuel must be handled—and, as shown next, from how many sites—to assemble even a single core.



Figure 4.2: Cumulative minor actinide delivered as a function of assemblies drawn, for three selection policies. All policies terminate at the same total, since they draw from one finite pool; they differ only in how quickly minor actinide accumulates. The dashed line marks one 27 t minor-actinide-only core; markers indicate where each policy first supplies a full core. **highest_minor_actinide** reaches a core with roughly 2.7 times fewer draws than **lowest_minor_actinide**.

4.2.2 Geographic Provenance of the Feed

Because the selection order is also a geographic order, each policy implies a different set of sites supplying the first core. Figure 4.3 shows the minor actinide contributed by each facility to the first 27 t core under the set A policies. The contrast is stark. The **first** policy, drawing in load order, concentrates the core in a handful of sites, each contributing on the order of 2 t; **highest_minor_actinide** instead spreads the draw across many facilities, tapping the minor-actinide-rich assemblies wherever they reside, so that no single site dominates. Ranking for feed quality thus trades a geographically concentrated retrieval for a distributed one—a consideration that connects directly to transportation and logistics, which fall outside the scope of this work but are enabled by the provenance recorded in `us_inventorySupply` (Table 3.2).

The effect is not specific to minor-actinide ranking. When the pool is instead ordered by discharge age (**older**, **newer**) or by reported burnup (**highest_burnup**, **lowest_burnup**), the set of sites supplying the first core shifts again, each policy favoring the facilities whose assemblies rank highest under its criterion. This confirms that the archetype’s selection acts as a general lever on feed provenance: any physical property

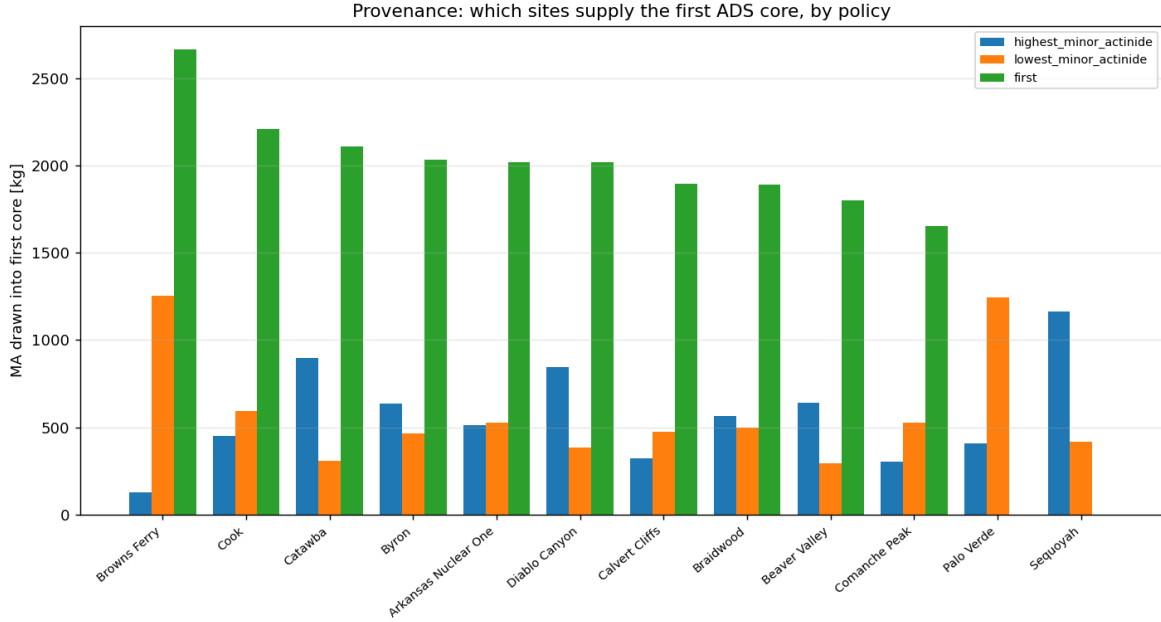


Figure 4.3: Minor actinide contributed by each facility to the first 27 t core, for the set A policies. **first** concentrates the core in a few sites drawn in load order, while **highest_minor_actinide** distributes it across many facilities.

recorded in the assembly data can be used to reshape both the composition and the geographic origin of the material entering the fuel cycle.

Finally, that the mechanism operates identically inside a full CYCLUS simulation—not only in offline analysis of the pool—was confirmed in a standalone `us_inventory` run in which the average composition of the supplied material tracked the active policy, with plutonium- and burnup-ranked policies delivering the correspondingly higher plutonium and fissile fractions expected from the ranking.

4.3 ADS Transmutation Performance

This section examines how the ADS transmutes the separated feed. Of the eight runs defined in Section 3.4.2, most coincide: as anticipated there, the once-through and recycling modes produce nearly identical histories wherever the burner is cycle-limited rather than feed-limited, so a single representative run suffices for those cases. The results therefore reduce to two comparisons that isolate the physics of interest. The first contrasts the two burner configurations—the minor-actinide-only core (ADS 1) and the 50/50 plutonium–minor-actinide core (ADS 2)—over the 100-year horizon, establishing how core size and feed composition govern the burner’s behavior. The second contrasts recycling against once-through operation for the minor-actinide-only core, the one pairing where the two modes diverge. Throughout, the per-cycle transmutation physics is identical between modes and horizons; what changes is how many cycles the burner completes before it exhausts its feed.

4.3.1 Per-Cycle Transmutation and Burn Fraction

At each discharge the core composition is read from the matched SERPENT case (Section ??), giving the surviving actinides and the fission products produced during the cycle. Figures 4.4 and 4.5 show the cumulative fission-product yield of the two cores, plotted against fission-product mass number. Both are the familiar double-humped curve of low-to-moderate-energy actinide fission, with a light peak near $A \approx 100$ and a heavy peak near $A \approx 138$ separated by the symmetric-fission valley. For the minor-actinide-only core (Figure 4.4) this signature appears even though the feed contains no uranium or plutonium, confirming that the minor actinides themselves are fissioning in the fast, subcritical core: the double hump is a property of the fission fragments’ shell structure, not of which actinide split. In both cases the three burnup snapshots nearly overlie one another, since the cumulative yield shape is set by the actinide mixture, which shifts only slowly with irradiation.

The two cores differ in the details of the distribution, and the difference reflects their feed. The 50/50 core (Figure 4.5) contains a large plutonium fraction, and ^{239}Pu fissions from a heavier compound nucleus than the minor-actinide vector; heavier fissioning systems place the light peak slightly higher in mass and fill in the symmetric valley, reducing the peak-to-valley ratio. The minor-actinide-only core, fissioning a lighter mix dominated by americium and neptunium, is consistent with a marginally deeper valley and a light peak shifted slightly lower. The shift is subtle because both feeds fission in the same fast spectrum and span overlapping actinide masses, but it is the expected signature of a plutonium-bearing versus a plutonium-free feed, and it confirms that the reduced-order model carries the compositional distinction through to the fission-product inventory.

The per-cycle mass accounting is summarized by the burn fraction, the ratio of actinide mass destroyed to actinide mass charged. For the minor-actinide-only core this fraction is constant at 41.3% across every cycle (Figure 4.6), a direct consequence of the reduced-order model: each core is irradiated to the same point on the same depletion trajectory, so each cycle removes the same fraction of its charge. For the 50/50 core the burn fraction rises to just under 50% (Figure 4.7); the plutonium half of that feed fissions more readily than the minor-actinide vector, so a larger fraction of each charge is consumed per pass.

The burn fraction reported here is the net actinide mass fissioned, obtained from the discharge mass balance (Equation 3.13). The raw `ActinideDestroyed` event (Equation 3.14) is not used as the performance metric, because for the minor-actinide-only core it overstates destruction: it attributes to “destroyed” the minor actinide that transmutes into plutonium, without crediting that bred plutonium, which remains in the core as actinide. The mass-balance quantity closes against the fission-product yield to within the fission mass defect (of order 0.1%) and is used throughout.

This per-cycle rate can be checked against first principles. Fissioning one kilogram of actinide releases approximately 0.93 GW-d of thermal energy, so a 3000 MW_{th} core destroys about 3.2 kg of actinide per full-power day, or roughly 1,120 kg yr⁻¹ at full power. Applied to the ten-year minor-actinide cycle, this predicts a per-cycle destruction close to the 11,150 kg read from the SERPENT case, agreeing to within a few percent. The reduced-order transmutation model is therefore consistent with the fission energy balance, an independent validation of the archetype’s mass accounting.

4.3.2 Core Size and Feed Composition: the Two Burner Cases

The two burner configurations produce very different operating histories, driven entirely by core size and the resulting cycle length. Figures 4.8 and 4.9 show the core state—irradiating or idle—over the 100-year horizon

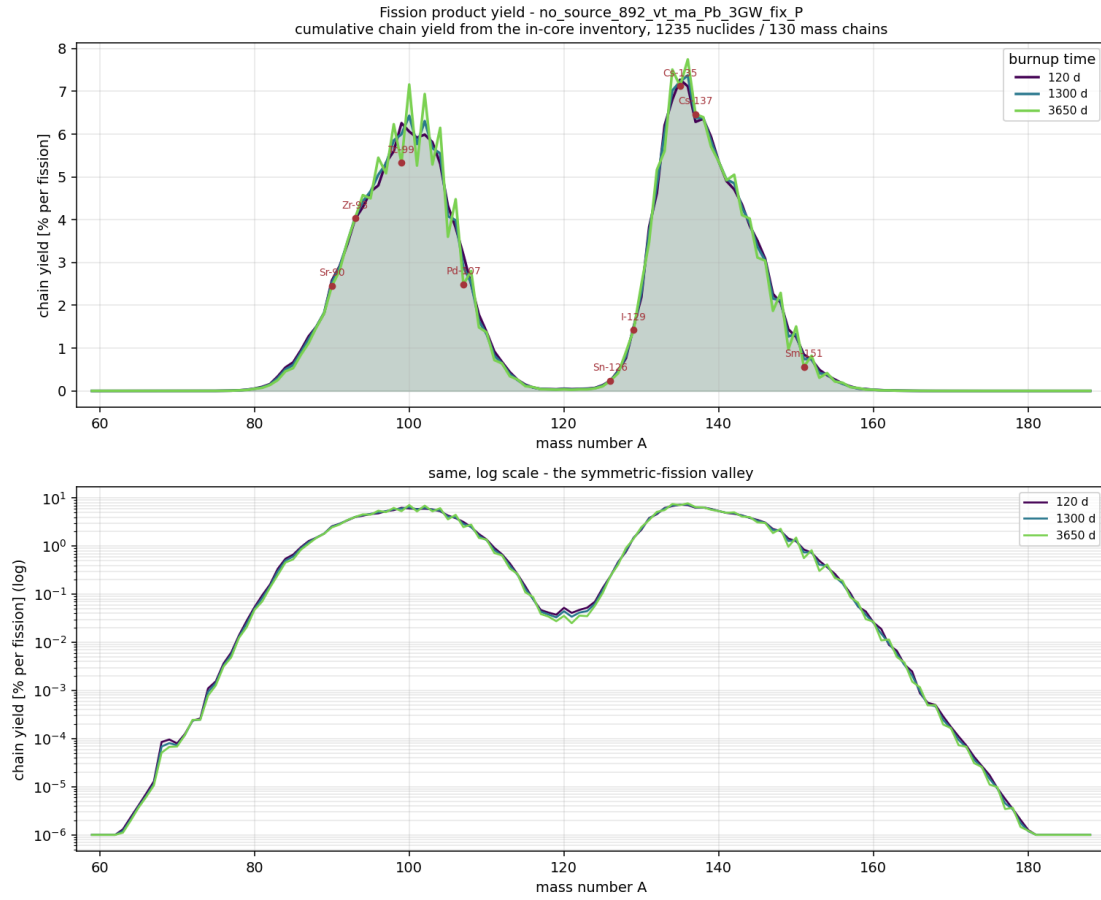


Figure 4.4: Cumulative fission-product chain yield of the minor-actinide-only core (ADS 1) as a function of fission-product mass number, at three points in the irradiation. The double-humped distribution appears despite the absence of uranium or plutonium in the feed, confirming that the minor actinides are the fissioning species.

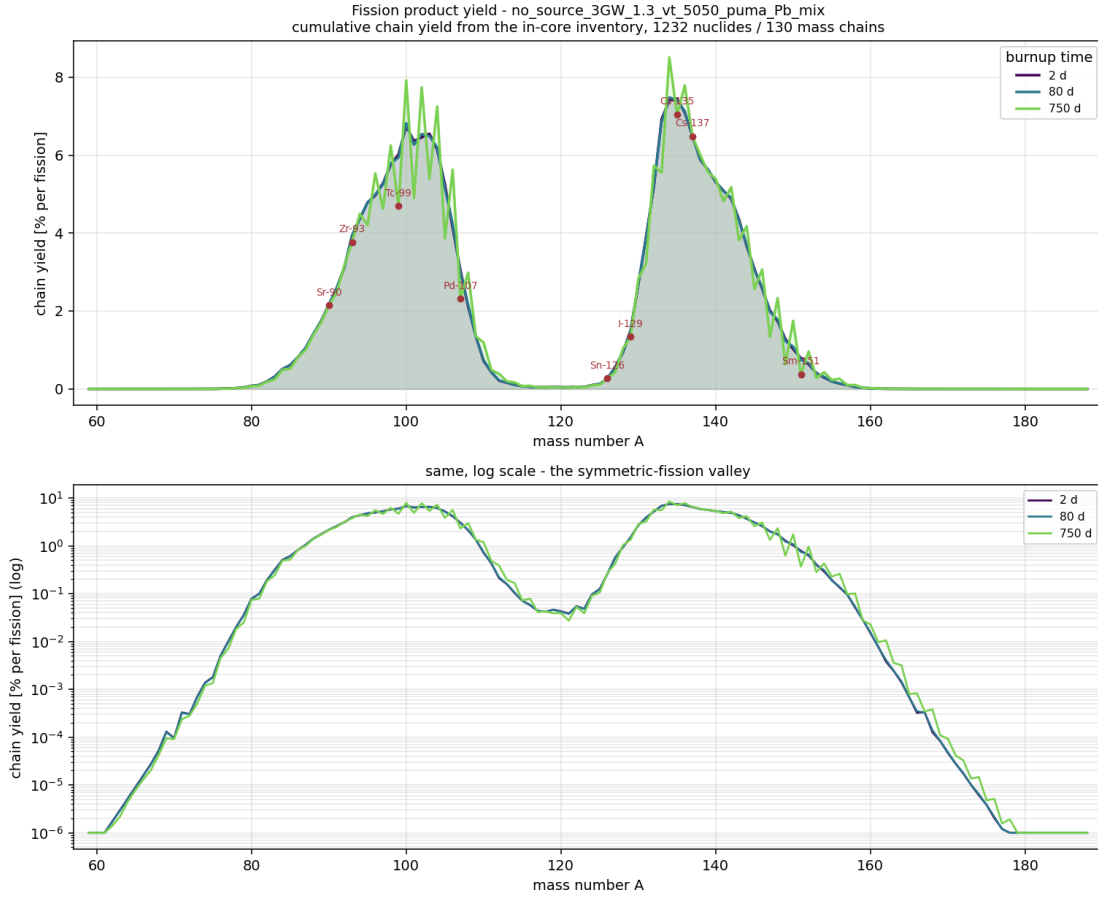


Figure 4.5: Cumulative fission-product chain yield of the 50/50 plutonium–minor-actinide core (ADS 2). Relative to the minor-actinide-only core, the plutonium-bearing feed fills in the symmetric-fission valley and shifts the light peak slightly higher in mass, the expected signature of a heavier fissioning system.

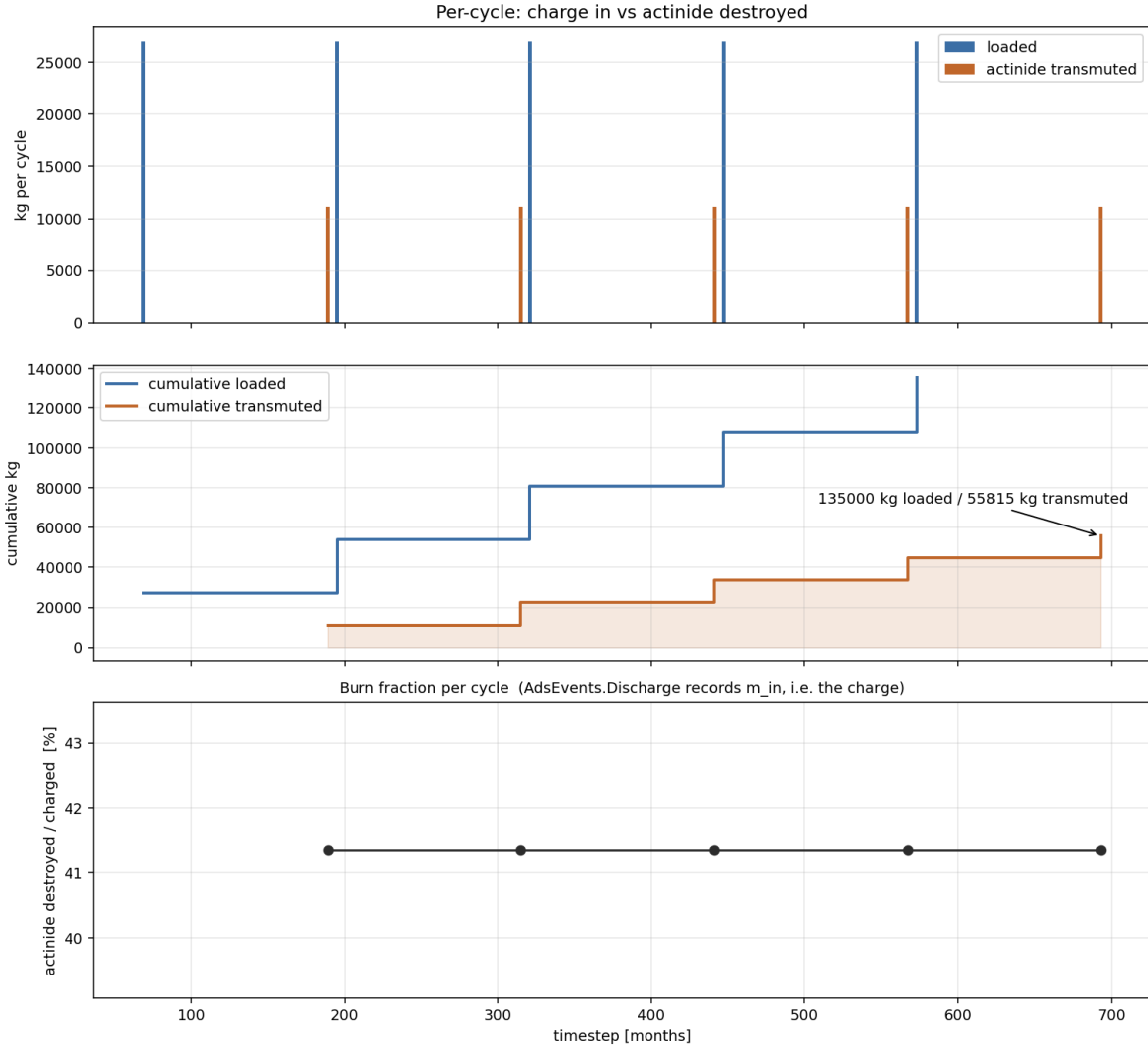


Figure 4.6: Per-cycle charge and actinide destruction for the minor-actinide-only core (ADS 1), 100-year recycling run. Each cycle charges 27 t and fissions 11.15 t, a constant burn fraction of 41.3% (bottom panel). The cumulative panel (centre) reaches 135 t charged and 55.8 t fissioned over five cycles.

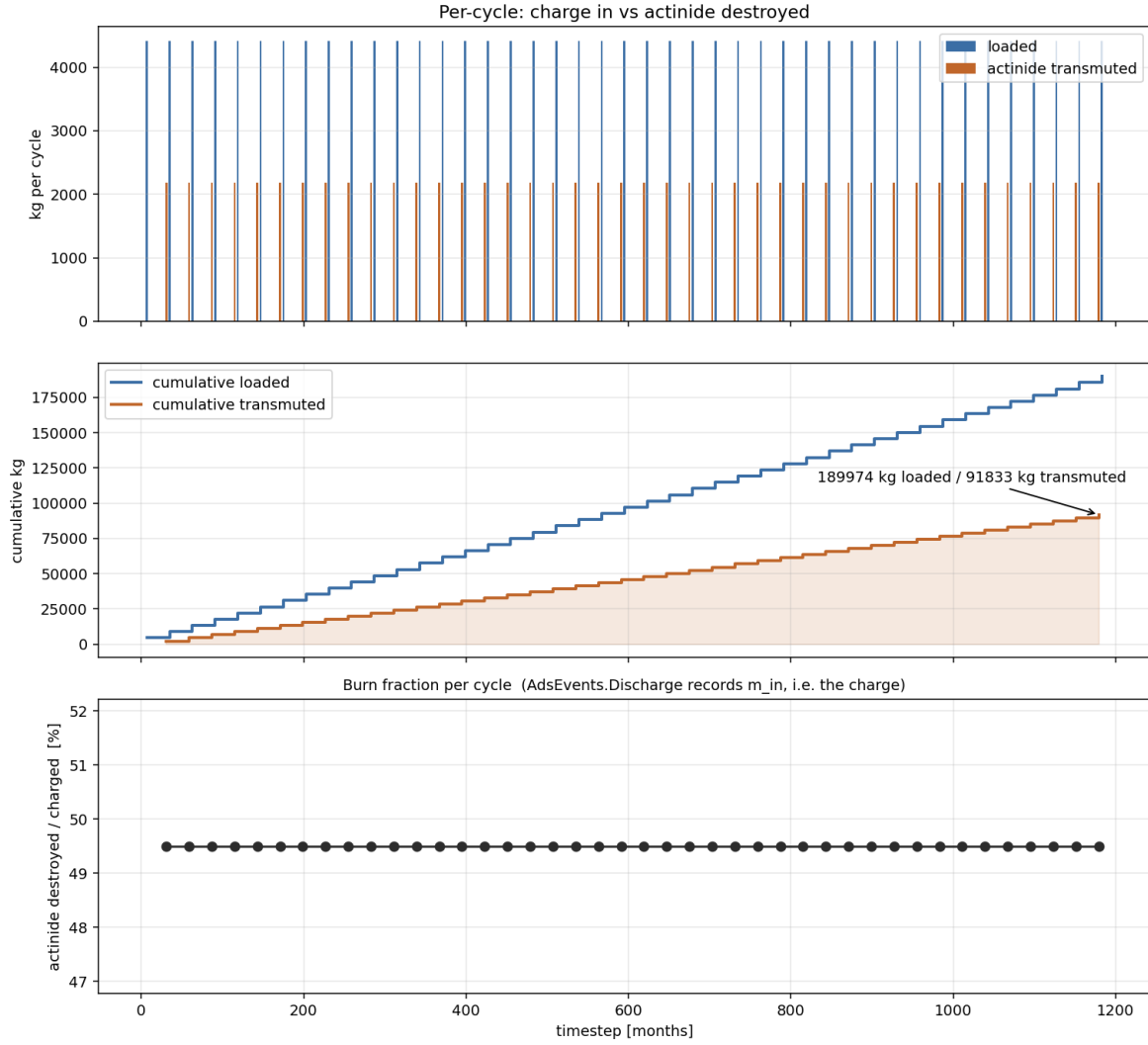


Figure 4.7: Per-cycle charge and actinide destruction for the 50/50 plutonium–minor-actinide core (ADS 2), 100-year recycling run. Each cycle charges 4.4 t and fissions 2.2 t, a burn fraction just under 50%. The small, fast-cycling core completes 42 cycles over the horizon.

for the two cases.

The minor-actinide-only core is large (27 t) and is irradiated for ten years per cycle. Over 100 years of recycling it completes only five cycles and then falls idle for the remainder of the horizon, reaching a duty factor of 50%. The core stops not because the horizon ends but because it runs out of feed: the accessible minor-actinide inventory is finite, and once it is consumed the burner has nothing to charge. The 50/50 core, by contrast, is small (4.4 t) and is cycled every two years. It completes 42 cycles and runs essentially continuously across the full horizon, at a duty factor of 85%. It never exhausts its feed because it consumes minor actinide slowly and supplements it with the far more abundant plutonium.

This is the central distinction between the two configurations. The minor-actinide-only core is *feed-limited*: its throughput is set by how much minor actinide the inventory can supply, not by reactor physics. The 50/50 core is *cycle-limited*: it processes feed as fast as its cycle allows, and the inventory is deep enough—because it draws on plutonium—to sustain that pace for a century. The trade-off is that the 50/50 core is no longer a pure minor-actinide burner: half of what it destroys is plutonium, so it addresses the broader transuranic inventory rather than the minor actinides specifically. Over the full horizon it destroys 91.8 t of actinide against the minor-actinide core’s 55.8 t, but that larger figure is achieved by co-burning plutonium the minor-actinide case leaves untouched.

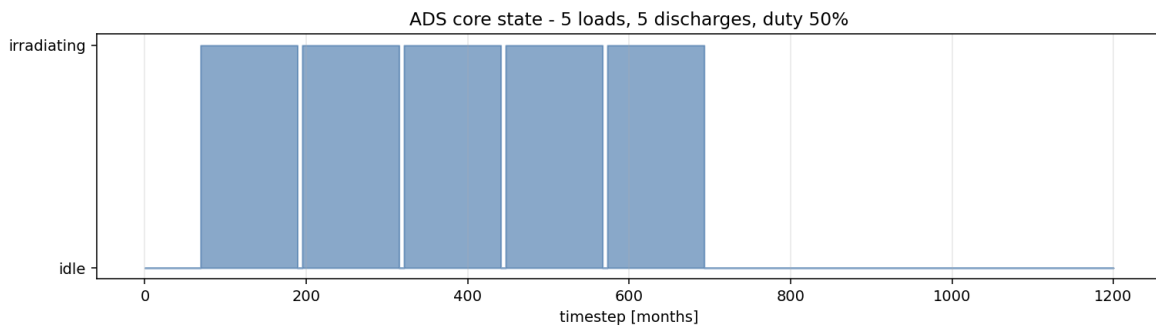


Figure 4.8: Irradiating-or-idle state of the minor-actinide-only core (ADS 1) over the 100-year recycling run. The core completes five ten-year cycles and then falls idle, having exhausted its feed; the duty factor is 50%.

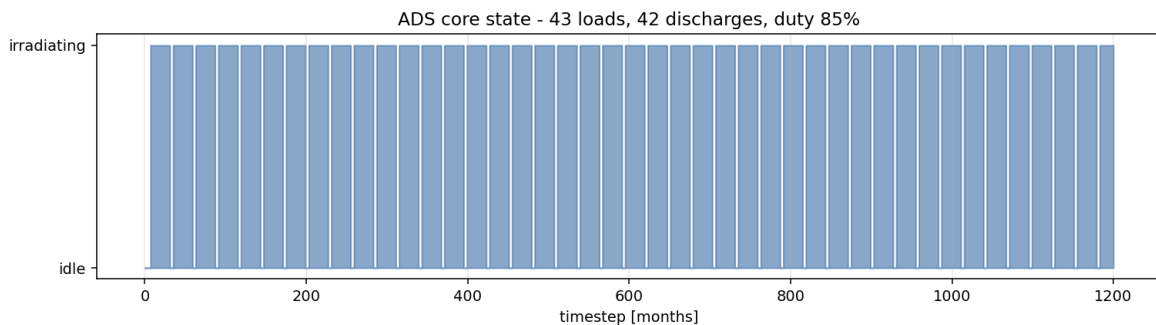


Figure 4.9: Irradiating-or-idle state of the 50/50 core (ADS 2) over the 100-year recycling run. The small, fast-cycling core runs essentially continuously at a duty factor of 85%, sustained by the plutonium fraction of its feed.

For the 50/50 core the once-through and recycling runs are indistinguishable, because the core is cycle-limited: within each two-year cycle there is little surviving actinide to recover before the next core is due, so

recycling changes almost nothing. This confirms the cycle-limited regime and is why only a single 50/50 run is shown.

4.3.3 Recycling versus Once-Through for the Minor-Actinide Core

The minor-actinide-only core is the one configuration where recycling materially changes the outcome, precisely because it is feed-limited. Comparing the recycling per-cycle history (Figure 4.6) with the once-through history (Figure 4.10) isolates the effect. The two runs share an identical per-cycle signature—the same 27 t charge, the same 11.15 t destroyed, and the same 41.3% burn fraction in the bottom panel of each figure—because the reduced-order depletion physics does not depend on where the feed came from. What differs is how many cycles the burner completes before its feed is spent: three under once-through, five under recycling.

The cumulative panels make the consequence concrete. In once-through mode the burner charges 81 t in total and fissions 33.5 t; with recycling it charges 135 t and fissions 55.8 t, an increase of $1.67\times$ in actinide destroyed. Recycling achieves this by recovering the surviving minor actinides from each discharge and returning them to the feed, so the same atoms make more than one pass through the core and the burner sustains operation past the point where the once-through supply is spent.

The charged figure is itself the clearest evidence of reuse. The total accessible minor-actinide inventory is only 103.8 t (Section 4.1), yet the recycling run charges 135 t—more minor actinide than exists in the national inventory. This is possible only because recycling returns material for additional passes; the excess of roughly 31 t is minor actinide irradiated more than once. The once-through run, by contrast, charges 81 t, comfortably within the inventory, and simply stops when the accessible feed is gone.

Even with recycling, however, the burner still stops well short of the horizon. The national minor-actinide inventory of 103.8 t is only about 3.8 core-loadings, so no amount of recycling can keep a 27 t core running indefinitely: recycling extends the campaign from three cores to five, not to the roughly nine a century of ten-year cycles could physically accommodate. The binding constraint is the inventory itself.

4.3.4 Transmutation Rate and Mission Scale

Taken together, these results frame the scale of the transmutation mission. A single 3000 MW_{th} unit destroys actinide at up to about 1,120 kg yr^{−1} at full power, reduced in proportion to its duty factor. Over a 30-year mission the 50/50 core, running near-continuously, destroys on the order of 27 t of actinide per unit, while the minor-actinide-only core destroys less because it stalls once its feed is spent. The two horizons reinforce this: a cycle-limited core scales its output roughly linearly with the length of the mission, whereas a feed-limited core plateaus once the accessible inventory is consumed, which is why extending the minor-actinide core from 30 to 100 years adds only two further cycles.

The consequence is that sizing a fleet to destroy the minor-actinide inventory is not the usual reactor-capacity problem. The entire national minor-actinide inventory is under four core-loadings; the limitation is how quickly that finite inventory can be retrieved, separated, and—critically—recycled, not how much reactor power is installed. Additional units shorten the calendar time to work through the available feed but cannot destroy more minor actinide than the inventory contains. A configuration that instead co-burns plutonium, as the 50/50 core does, sidesteps the feed limit but redefines the mission from minor-actinide destruction to bulk transuranic destruction. This inversion of the usual sizing question—feed-bound rather than power-bound—is the central practical finding of the transmutation results, and it is examined further in

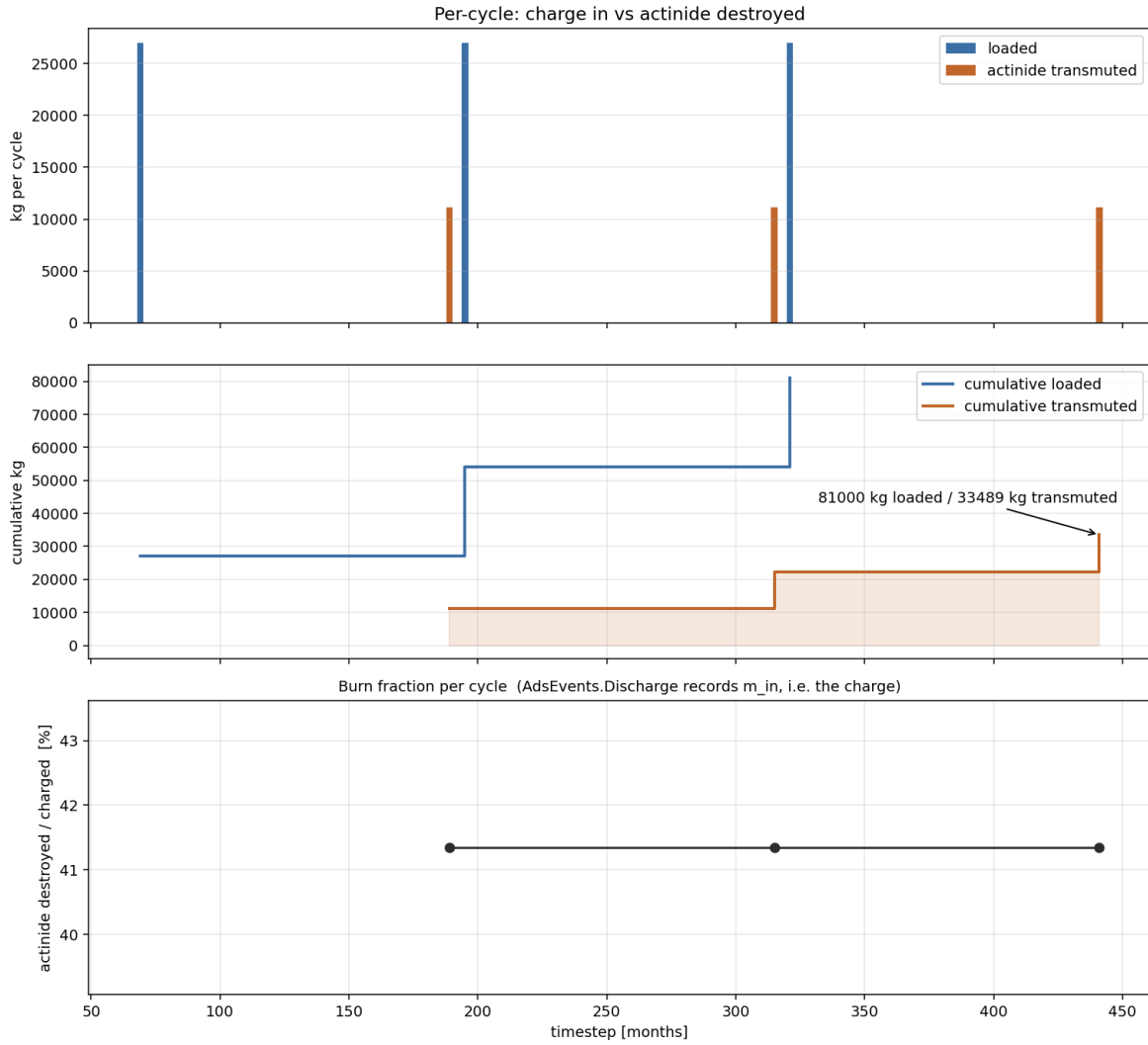


Figure 4.10: Per-cycle charge and actinide destruction for the minor-actinide-only core (ADS 1), 100-year once-through run. The per-cycle charge, destruction, and 41.3% burn fraction match the recycling run (Figure 4.6), but the burner completes only three cycles, reaching 81 t charged and 33.5 t fissioned before its feed is exhausted.

the discussion of Section 4.5.

4.4 Processed Inventory

The transmutation results of Section 4.3 describe what the burner does per cycle; this section follows the material through the whole fuel cycle and asks what becomes of the national inventory once it is processed. Every kilogram drawn from `us_inventory` meets one of three fates: it is destroyed by fission in the ADS, it is recovered and returned to the feed under recycling, or it is routed to disposal. Tracking these streams shows both what the mission accomplishes and what it leaves behind, and connects the results back to the waste-management motivation of Chapter 2.

4.4.1 Disposition of the Minor-Actinide Inventory

Table 4.1 accounts for the 103.8 t of accessible minor actinide (Section 4.1) at the end of the 100-year minor-actinide-only runs. In once-through operation the burner fissions 33.5 t—about a third of the national inventory—and leaves the remaining 70.3 t undestroyed. Of that remainder, roughly 47.5 t is surviving actinide discharged to waste, and about 22.8 t is never processed at all: once three cores have been drawn, the minor actinide left in the unprocessed inventory is less than a single 27 t core loading, so the burner cannot assemble a fourth core and stalls with that material stranded.

Recycling improves the destroyed fraction to 55.8 t, a little over half the national inventory, by returning surviving minor actinides for further passes rather than discarding them, and by drawing down essentially all of the accessible inventory rather than stranding a core’s worth. Even so, just under half of the minor actinide remains undestroyed at the end of the century, consistent with the feed-limited behavior established in Section 4.3.3: the 27 t core is simply too large relative to the available inventory to consume it within the horizon.

Table 4.1: Disposition of the 103.8 t accessible minor-actinide inventory at the end of the 100-year minor-actinide-only runs. “Remaining” combines surviving actinide sent to waste, material held in buffers, and, for once-through, minor actinide stranded in unprocessed inventory.

Disposition	Once-through	Recycle
Actinide destroyed (fissioned)	33.5 t	55.8 t
Actinide remaining	70.3 t	48.0 t
National inventory	103.8 t	103.8 t

4.4.2 Composition of the Waste Stream

Figure 4.11 shows the final composition of the waste repository for the minor-actinide-only recycling run, as the mass of each of the 25 most abundant nuclides. The stream is overwhelmingly uranium: ^{238}U alone accounts for roughly 8.5×10^4 t, four orders of magnitude above any minor actinide. This is a direct consequence of the scarcity documented in Section 4.1. Because minor actinide is only about 0.12% of the spent fuel, extracting the roughly 100 t the burner consumes requires separations to process on the order of the entire drawn inventory—tens of thousands of tonnes of spent fuel—and the uranium and fission products stripped from that fuel dominate the disposal stream. The mass sent to disposal is therefore not reduced by the transmutation mission; it is only changed in character.

That change in character is nonetheless the intended effect, and it is visible in the figure. Minor actinides are almost absent from the waste: americium, the most abundant minor actinide in the original fuel, does not appear among the major waste nuclides at all under recycling, having been recovered and burned. In the once-through run the same plot shows a small residual ^{241}Am peak, which recycling removes—a direct illustration of recovery closing that pathway. The minor-actinide-only cycle thus succeeds in stripping minor actinide from the waste, but does not reduce its volume.

The figure also exposes the principal limitation of the minor-actinide-only configuration. Plutonium is the second-largest component of the waste, with ^{239}Pu near 7×10^2 t: because this burner consumes only minor actinide, separations routes the entire plutonium content of every processed assembly straight to disposal, untouched. A cycle designed to destroy minor actinide therefore leaves the larger and equally long-lived plutonium inventory in the waste, which is precisely the gap the 50/50 configuration is meant to address.

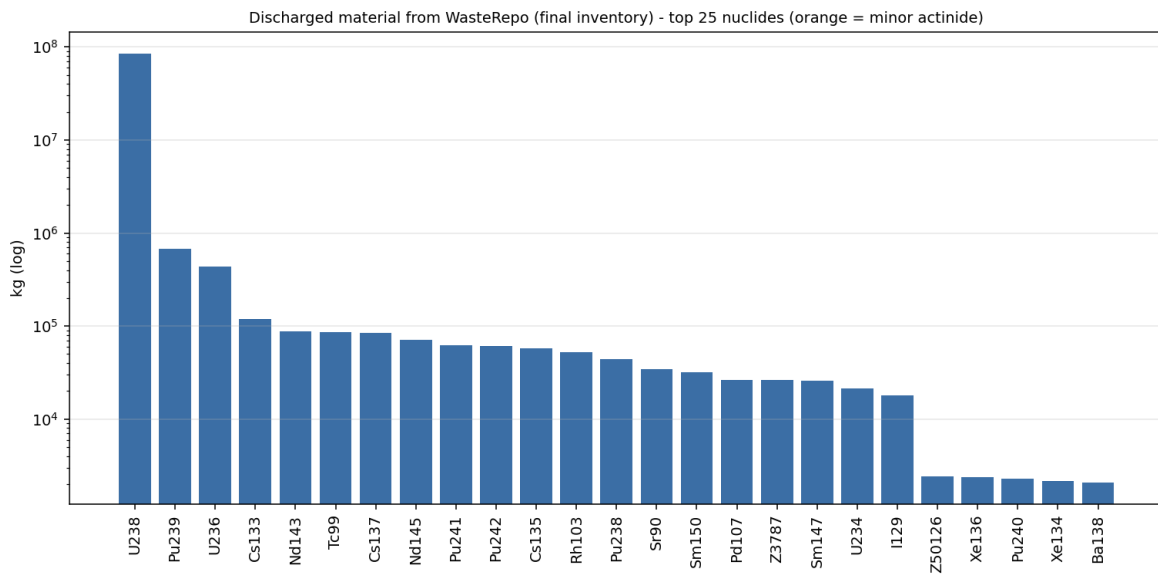


Figure 4.11: Final waste-repository composition for the minor-actinide-only recycling run (Case 3), showing the 25 most abundant nuclides on a logarithmic mass scale; minor actinides are highlighted. The stream is dominated by uranium and plutonium stripped by separations, while minor actinides are nearly absent, having been recovered and burned.

4.4.3 Material Recovered under Recycling

Where the minor-actinide-only cycle sends plutonium to waste, the 50/50 configuration returns it to the feed. Figure 4.12 shows the material discharged from the ADS in the 50/50 recycling run, which separations recovers and blends back into fresh feed. The stream is led by plutonium isotopes—surviving and bred ^{239}Pu , ^{238}Pu , and ^{240}Pu —followed by the minor actinides ^{237}Np , ^{244}Cm , ^{241}Am , and their heavier companions, alongside the fission products of the cycle. By recovering the plutonium as well as the minor actinides, the 50/50 cycle keeps that plutonium in the burner over successive passes rather than stranding it in waste, which is why it addresses the broader transuranic inventory rather than the minor actinides alone. The presence of $^{242\text{m}}\text{Am}$ in the recovered stream also confirms that the archetype preserves the metastable state through the composition transformation, rather than collapsing it onto the ground state.

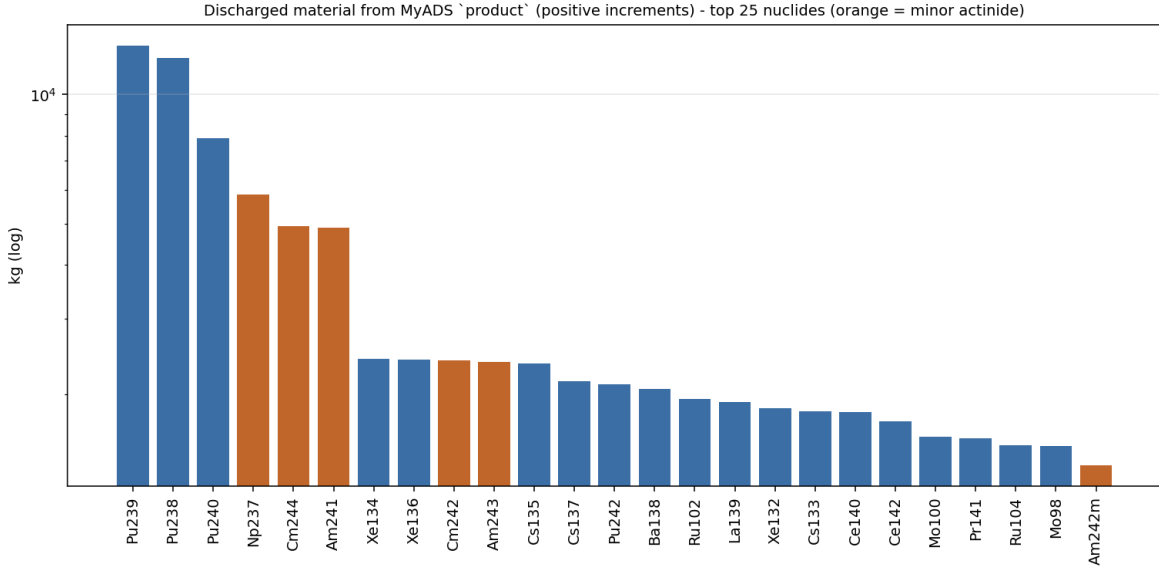


Figure 4.12: Composition of the material discharged from the ADS in the 50/50 recycling run (Case 4), showing the 25 most abundant nuclides on a logarithmic mass scale; minor actinides are highlighted. Plutonium leads the recovered stream, followed by the minor actinides and fission products, all of which recycling returns to the feed.

4.4.4 Implications for the Waste Inventory

Taken together, these results qualify what transmutation achieves for the national inventory. Over a century, even the more favorable recycling configuration destroys only about half of the accessible minor actinide, and it does so at the cost of processing essentially the entire spent-fuel inventory, the bulk of which— uranium, fission products, and, for the minor-actinide-only cycle, plutonium— still requires disposal. The mission changes the isotopic character of the waste by removing minor actinide, which is the component that dominates its long-term radiotoxicity and heat load; it does not reduce the mass or volume requiring a repository. Quantifying the resulting reduction in radiotoxicity or decay heat would require coupling these mass histories to a decay and dose calculation, which is beyond the scope of this work but is a natural extension of it (Chapter 6). What the fuel-cycle results establish is the upstream reality any such benefit must be weighed against: minor-actinide transmutation is constrained not by reactor performance but by the scarcity of its feed and the scale of the inventory that must be handled to extract it.

4.5 Discussion

The results of this chapter can be read as a single argument about scale. The `us_inventory` and `ads` archetypes were built to connect a detailed, assembly-resolved picture of the U.S. spent-fuel inventory to a reduced-order model of an accelerator-driven burner, and running them together turns qualitative expectations about minor-actinide transmutation into quantified constraints. Two findings dominate, and both point the same way.

4.5.1 The Mission is Feed-Bound, Not Power-Bound

The central result is that a minor-actinide transmutation mission is limited by the availability of its feed rather than by installed reactor capacity. The accessible minor-actinide inventory is 103.8 t—under four loadings of a single 27 t core—so the binding question is not how much power can be deployed but how quickly that finite, dilute inventory can be retrieved, separated, and recycled. The minor-actinide-only core makes this concrete: it is feed-limited, stalling after three cores once-through or five with recycling (Section 4.3.3), and no increase in reactor power can change that, because the feed to sustain further cores does not exist. This inverts the usual framing of a reactor deployment, in which capacity is the constraint and fuel is assumed abundant.

A simple rate calculation frames the fleet question that follows from this. A 3000 MW_{th} unit destroys actinide at up to about 1,120 kg yr⁻¹ at full power (Section 4.3.4), so clearing the 103.8 t inventory represents on the order of 90 unit-years of full-power operation. Distributed over a 30-year mission this is roughly three units' worth of destruction capacity; a single unit would require most of a century, and the feed-limited minor-actinide core cannot in practice run at full duty for that long. This is not a precise fleet-sizing—it omits reprocessing losses, retrieval and transport logistics, and the duty-factor penalty the feed limit itself imposes—but it establishes the order of magnitude: the transmutation of the national minor-actinide inventory is a task for a handful of high-power units over decades, and its pace is set by fuel handling rather than by how many units are built.

The 50/50 configuration escapes the feed limit, but only by changing the mission. By co-burning plutonium—ten times more abundant in the inventory than minor actinide—the small, fast-cycling core stays fed for the full century at a 85% duty factor and destroys 91.8 t of actinide, far more than the minor-actinide core's 55.8 t (Section 4.3.2). But roughly half of that is plutonium, so the 50/50 core is a transuranic burner rather than a minor-actinide burner. The choice between the two configurations is therefore a choice of mission: destroy the scarce, high-radiotoxicity minor actinides at a pace bounded by their scarcity, or destroy the broader transuranic inventory at a higher rate by accepting plutonium as feed.

4.5.2 Transmutation Changes the Waste's Character, Not its Volume

The second finding concerns what the mission does to the waste. Because minor actinide is only 0.12% of the spent fuel, extracting the roughly 100 t the burner consumes requires separations to process on the order of the entire drawn inventory, and the uranium, fission products, and—for the minor-actinide-only cycle—plutonium stripped from that fuel dominate the disposal stream (Section 4.4.2). The transmutation mission thus does not reduce the mass or volume destined for a repository. What it changes is the waste's isotopic character: it removes minor actinide, the component that dominates the long-term radiotoxicity and decay heat of spent fuel. That is a meaningful outcome for repository performance, but it is a qualitative change to the waste rather than a reduction of it, and it is bought at the cost of handling the full inventory.

This reframes the value proposition of transmutation in a way the fuel-cycle model is well suited to expose. The benefit is not less waste but longer-lived constituents removed from it; the cost is a fuel cycle that must retrieve, transport, and reprocess essentially the entire national inventory to recover a trace component. Whether that trade is worthwhile is a question about repository requirements and reprocessing capacity, not about the burner itself.

4.5.3 Limitations

Several limitations bound the interpretation of these results, and identifying them also marks the most useful directions for further work.

The ADS model is reduced-order. The burner does not solve neutron transport during the simulation; it reads precomputed SERPENT depletion cases and selects and interpolates among them (Section 3.3). This is appropriate for a fuel-cycle study, where the quantity of interest is the mass transmuted per core rather than the detailed in-core physics, and the per-cycle rate was shown to agree with an independent fission-energy balance to within a few percent (Section 4.3.1). But it inherits the assumptions of the underlying cases—a single homogeneous fuel region, a fixed geometry, and a library spanning only the two representative feeds used here—and it cannot capture behavior outside the tabulated cases. A core whose composition drifts far from any library case is matched to the nearest one rather than modeled directly, and the constant per-cycle burn fraction is a consequence of irradiating every core to the same point on the same trajectory, not a physical invariant.

The inventory-selection results characterize the feed directly from the static inventory rather than from the coupled simulation (Section 4.2). This is exact, because the selection is a deterministic ordering of a fixed pool, but it means the policy comparison isolates the supply side and does not reflect any feedback between selection and burner operation. All production scenarios use the single `highest_minor_actinide` policy, so the effect of policy on full-cycle outcomes, as opposed to on the feed, is left for future work.

Finally, the study stops at the mass balance. It quantifies what is destroyed and what remains but does not evaluate the radiological or economic consequences that determine whether the mission is worthwhile. Radioactive decay is modeled within the simulation, so material compositions are aged as they move through the fuel cycle; but no downstream radiotoxicity or decay-heat calculation is applied to the final waste compositions, so the reduction in long-term radiological hazard that motivates minor-actinide removal is not quantified here. Likewise, no cost, facility, or transportation model accompanies the fuel cycle, so the substantial reprocessing and handling burden the results imply is described but not priced. Nor does the model represent the retrieval and transport logistics that the provenance results (Section 4.2.2) are designed to feed. These are not incidental omissions but the natural next steps: the assembly-resolved provenance, the mass-conserving discharge streams, and the feed-limited operating histories developed here are precisely the inputs a subsequent radiotoxicity, economic, or siting analysis would require, and building those couplings onto this framework is the direction the concluding chapter identifies (Chapter 6).

Chapter 5

Conclusions

Chapter 6

Conclusions and Future Work

This thesis set out to connect a detailed picture of the U.S. used-fuel inventory to a model of an accelerator-driven burner, so that the transmutation of minor actinides could be evaluated as a national fuel cycle rather than as an isolated reactor calculation. Two Cyclus archetypes were developed to that end. `us_inventory` represents the inventory at assembly resolution, supplying material by a choice of selection policies while preserving each assembly’s composition and provenance; `ads` models an accelerator-driven minor-actinide burner from precomputed Serpent depletion cases, transforming a loaded core into its irradiated composition and closing the actinide mass balance each cycle. Together they form a complete transmutation fuel cycle when combined with standard separations, mixing, and disposal facilities.

Exercising this fuel cycle across eight scenarios produced a consistent picture, summarized by two findings. First, minor-actinide transmutation is bounded by its feed rather than by reactor capacity. The accessible minor-actinide inventory is only about 104 t—under four core-loadings—so the large minor-actinide-only burner exhausts its supply and stalls, destroying about a third of the inventory once-through and just over half with recycling, while no increase in reactor power can extend it further. The smaller fifty-fifty core avoids this limit only by co-burning plutonium, which changes the mission from minor-actinide destruction to broader transuranic destruction. Second, because minor actinide is a trace component of the spent fuel, the mission alters the character of the waste—removing the constituents that dominate its long-term hazard—without reducing the mass or volume that requires disposal, and it does so at the cost of reprocessing essentially the entire drawn inventory. The per-cycle transmutation rate was shown to agree with an independent fission-energy balance to within a few percent, giving confidence in the reduced-order model.

These findings temper the expectation that accelerator-driven transmutation offers a straightforward reduction of the waste burden. It can strip the most hazardous component from the waste, but it is a decades-long undertaking for a small fleet of high-power units, limited by fuel handling rather than reactor performance, and it leaves the bulk of the disposal problem intact.

Several extensions follow directly from this work. The most immediate is to couple the mass-conserving waste histories to a decay and dose calculation, so that the reduction in long-term radiotoxicity and decay heat—the benefit that motivates minor-actinide removal—can be quantified rather than inferred. A second is to use the assembly-level provenance recorded by `us_inventory`, which associates each supplied assembly with its source facility and coordinates, as the input to a transportation and siting analysis of where retrieval and processing would occur. A third is an economic model of the reprocessing and handling burden the results imply, which this study describes but does not price. Finally, the reduced-order `ads` model could be

broadened with a larger library of depletion cases, or the selection-policy effect examined within the coupled simulation rather than on the static inventory alone. In each case, the archetypes and results developed here are intended as the foundation such analyses would build upon.

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Appendix A

An appendix

Appendix.